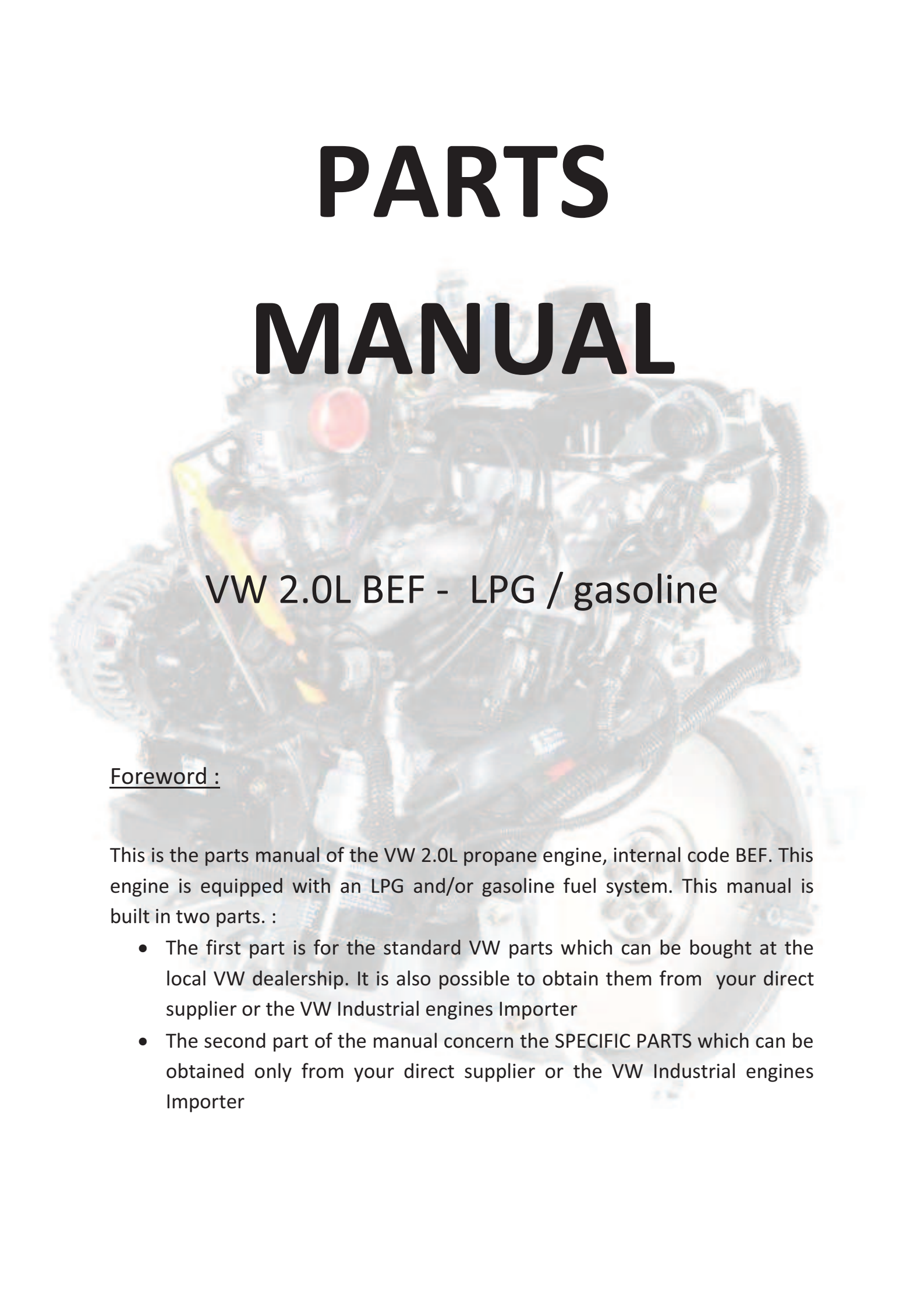
PARTS MANUAL

VW 2.0L BEF - LPG / gasoline



Foreword :

This is the parts manual of the VW 2.0L propane engine, internal code BEF. This engine is equipped with an LPG and/or gasoline fuel system. This manual is built in two parts. :

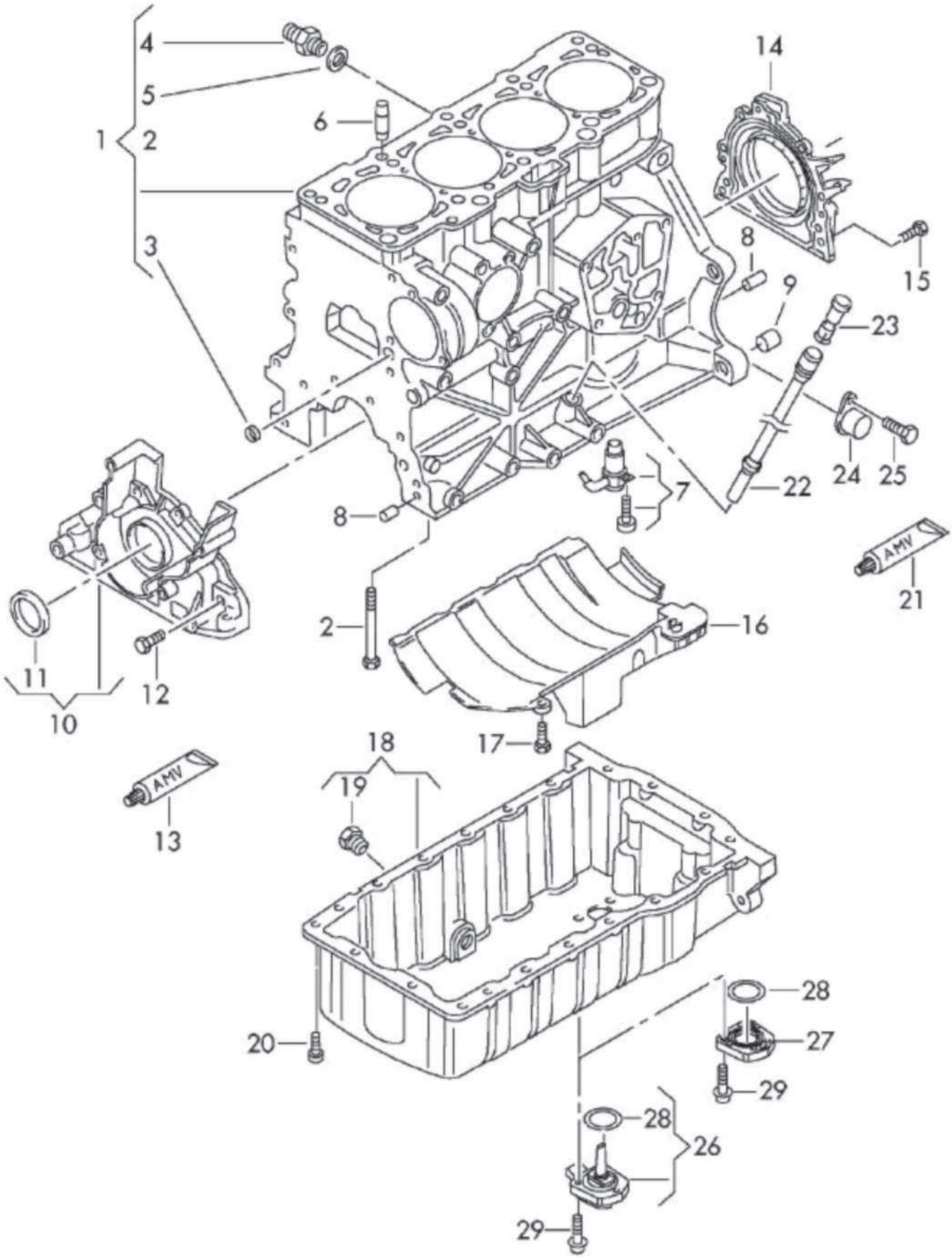
- The first part is for the standard VW parts which can be bought at the local VW dealership. It is also possible to obtain them from your direct supplier or the VW Industrial engines Importer
- The second part of the manual concern the SPECIFIC PARTS which can be obtained only from your direct supplier or the VW Industrial engines Importer

Table of contents

1. VW parts 3
1.1. Cylinder block / oil sump 3
1.2. Crankshaft / conrod / bearings 5
1.3. Pistons / piston rings 7
1.4. Cylinder head 9
1.5. Camshaft / valves 11
1.6. Toothed belt / toothed belt guard 13
1.7. Oil pump / oil dipstick 15
1.8. Oil filter bracket 17
1.9. Coolant pump 19
1.10. Coolant pipe 21
1.11. Exhaust manifold 23
1.12. Alternator + mounting parts 25
1.13. Ignition 27
2. SPECIFIC PARTS 29
2.1. Pressure regulator + mounting parts 31
2.2. Intake module + mounting parts 33
2.3. Gas mixer + fuel control valve 35
2.4. ECU + mounting parts 37
2.5. Engine brackets + mounting parts 39
2.6. Flywheel housing + mounting parts 41
2.7. Harness + mounting parts 43
2.8. Starter + mounting parts 45
2.9. Water hoses + mounting parts 47
2.10. Gasoline fuel parts 49

1. VW parts

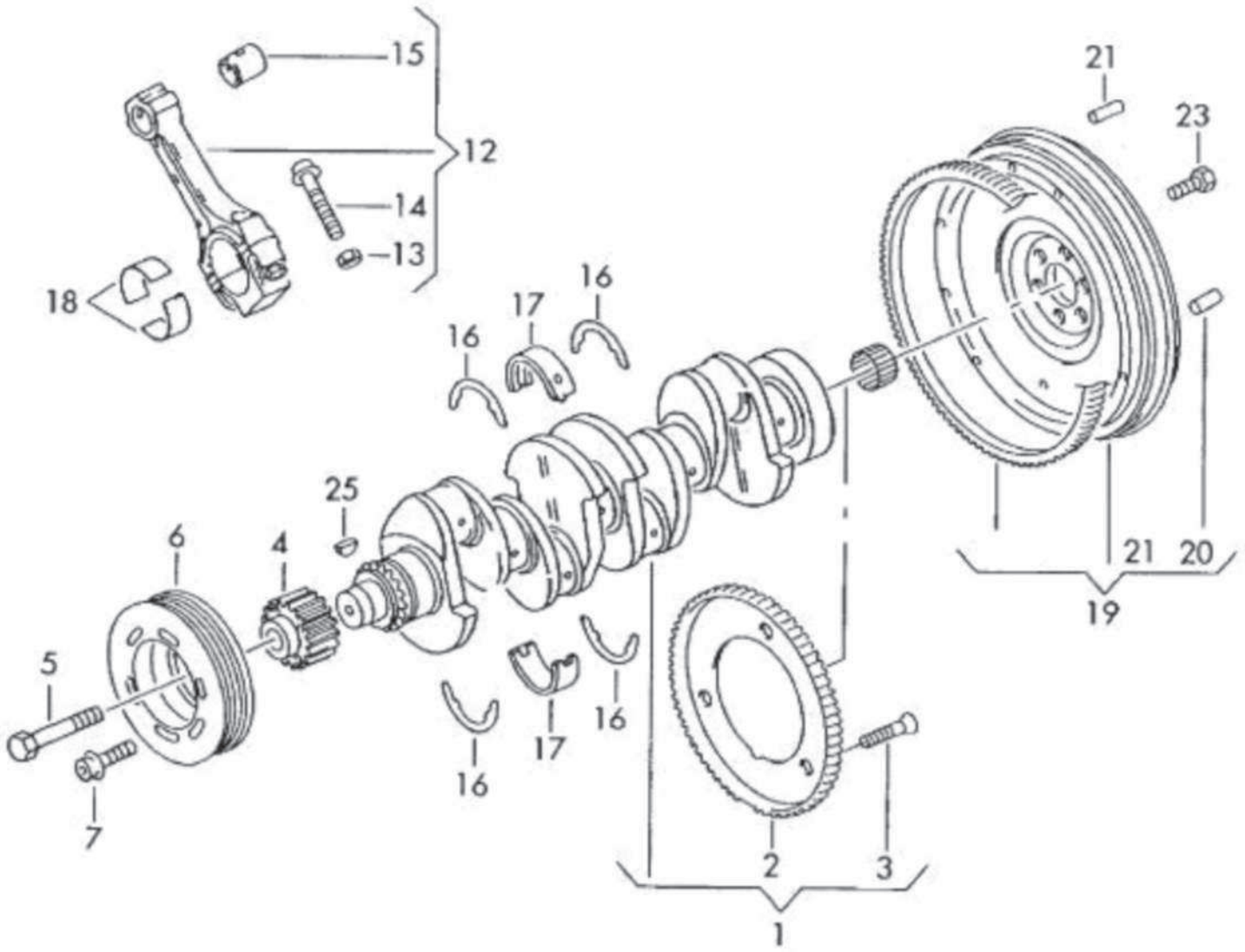
1.1. Cylinder block / oil sump



cylinder block / oil sump

POS	PART NUMBER	DESCRIPTION	QTY
1	06A 103 101 AE	cylinder block with pistons	1
2	N 901 300 01	hexagon bolt M10X80	10
3	N 011 907 10	end cap	3
6	037 103 139	cylinder pin 8M6X20	1
6	037 103 139 A	cylinder pin 4M6X20	1
7	06B 103 154	oil spray jet	4
8	N 043 207 2	cylinder pin 8X14	4
9	037 103 139	cylinder pin 8MM	1
9	037 103 139 A	cylinder pin 4MM	1
9	014 301 153	dowel sleeve 16X12.8	2
10	038 103 151 H	sealing flange front	1
11	038 103 085 E	shaft oil seal 35X48X10	1
12	N 909 450 02	hexagon socket head bolt with shoulder M7X21.5	6
13	D 176 404 A2	silicone sealant sealing flange	X
14	06A 103 171 A	sealing flange rear	1
15	N 909 450 02	hexagon socket head bolt with shoulder M7X21.5	6
16	06B 103 623 P	surge lock	1
17	N 907 040 07	hexagon socket head bolt with shoulder M7X55	1
18	038 103 603 N	engine oil sump with opening for oil level sensor	1
19	N 908 132 02	seal bolt with sealing ring	1
20	N 907 356 04	hexagon socket head bolt with shoulder	20
21	D 176 404 A2	silicone sealant oil sump	X
22	06D 103 634 E	tube for oil dipstick	1
23	06A 103 663 C	guide funnel	1
24	078 906 433 B	impulse sender	1
25	N 909 450 02	hexagon socket head bolt with shoulder M7X21.5	1
26	1J0 907 660 C	oil level sensor	1
27	03G 103 707	cover for oil sump	1
28	038 103 196	o-ring	1
29	N 910 652 01	socket head collared bolt with inner multipoint head	3

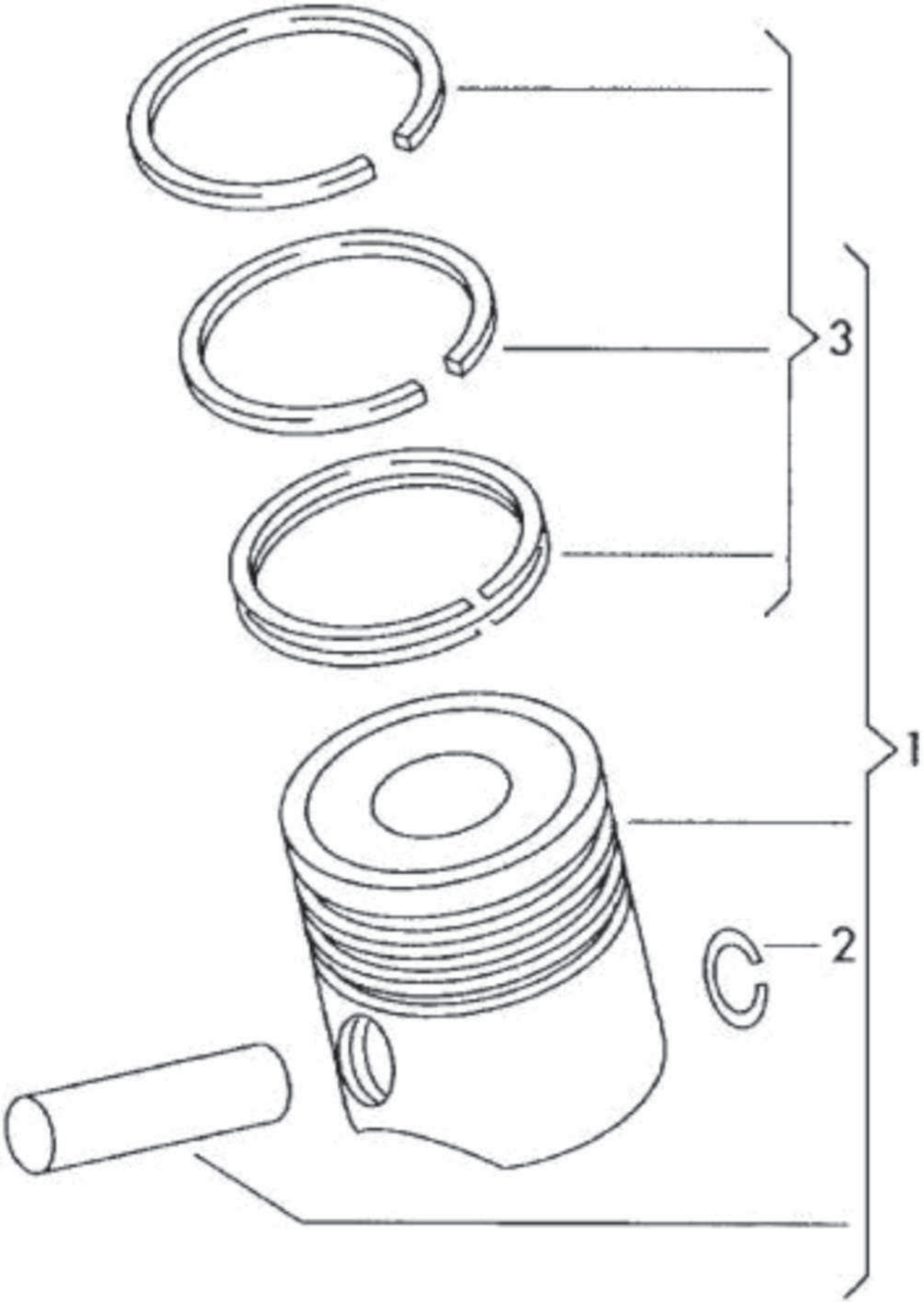
1.2. Crankshaft / conrod / bearings



crankshaft / conrod / bearings

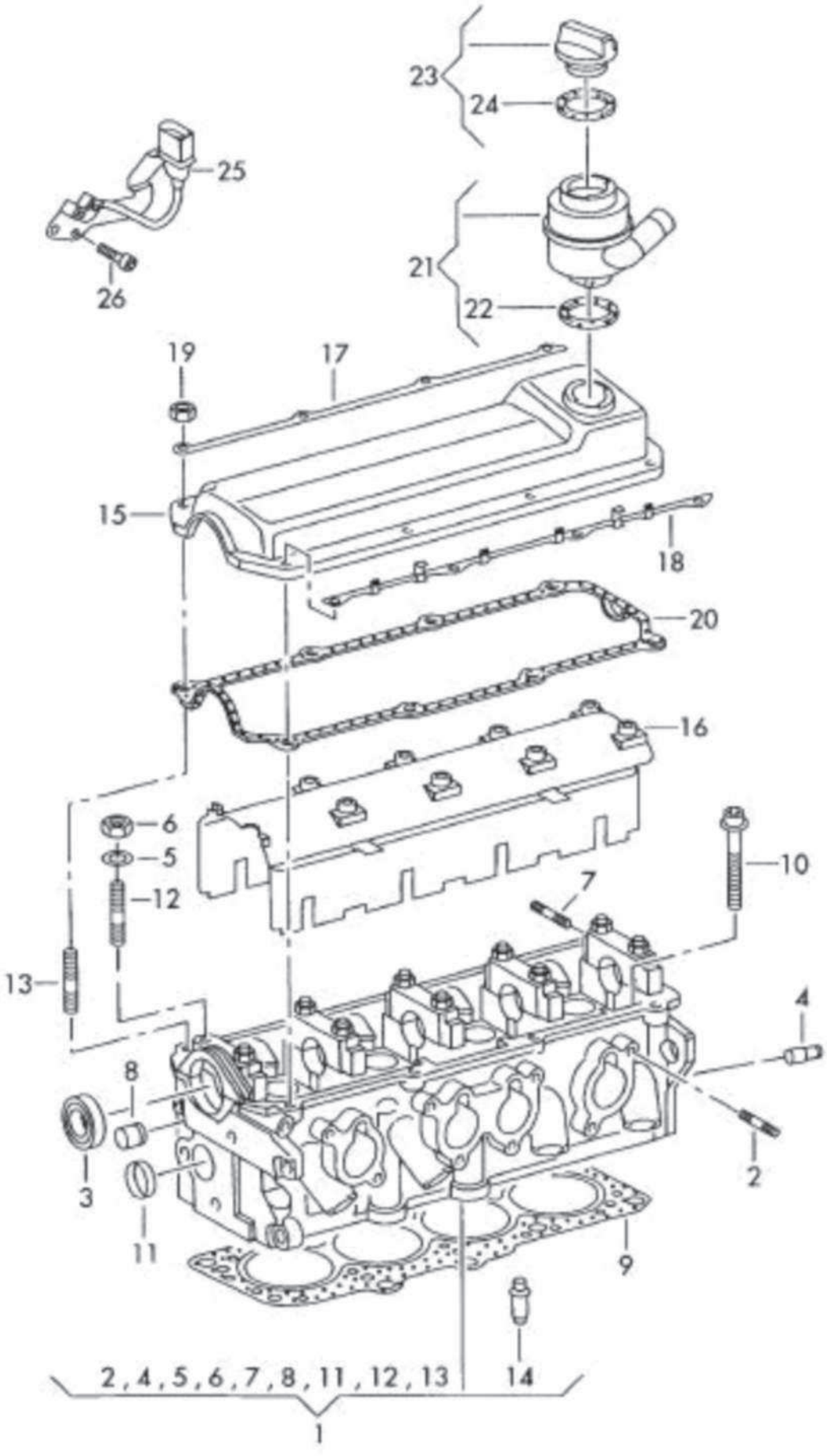
POS	PART NUMBER	DESCRIPTION	QTY
1	06A 105 021 G	crankshaft	1
2	050 105 189 B	impulse rotor	1
3	053 105 229 A	countersunk bolt M6X16	3
4	06A 105 263 E	toothed belt pulley	1
5	WHT 005 322	12-point head bolt M16X1.5X54	1
6	06A 105 243 E	vibration damper	1
7	N 902 428 04	hexagon socket head bolt M8X18	4
8	048 105 323 B	clutch plate	1
9	051 105 301	washer 0.5	1
10	056 105 303	compensation washer 1.2MM	X
11	N 909 462 01	12-point head bolt M10X1X17X10	6
12	027 198 401 D	1 set: connecting rods	1
13	048 105 427	hexagon nut	8
14	048 105 425	big end bolt press-fitted (rotation-angle controlled)	8
15	053 105 431 B	bush - connecting rod	4
16	026 198 421	1 set of thrust washers bearing 3	1
17	06B 105 561	crankshaft bearing shell upper bearing 1-5	5
17	06B 105 591	crankshaft bearing shell lower bearing 1-5	5
18	034 105 701	connecting rod bearing shell	8
19	026 105 269 L	flywheel	1
20	N 013 210 2	cylinder pin 6X16	3
21	N 101 168 01	cylinder pin 5X38	2
23	N 902 061 03	hexagon bolt M10X1X19.5	6

1.3. Pistons / piston rings



pistons / piston rings			
POS	**PART NUMBER**	**DESCRIPTION**	**QTY**
1	06A 107 065 BG	piston complete 82.5	4
2	N 012 273 1	securing ring 20X1.5	8
3	06A 198 151 C	1 set: piston rings 82.5	4

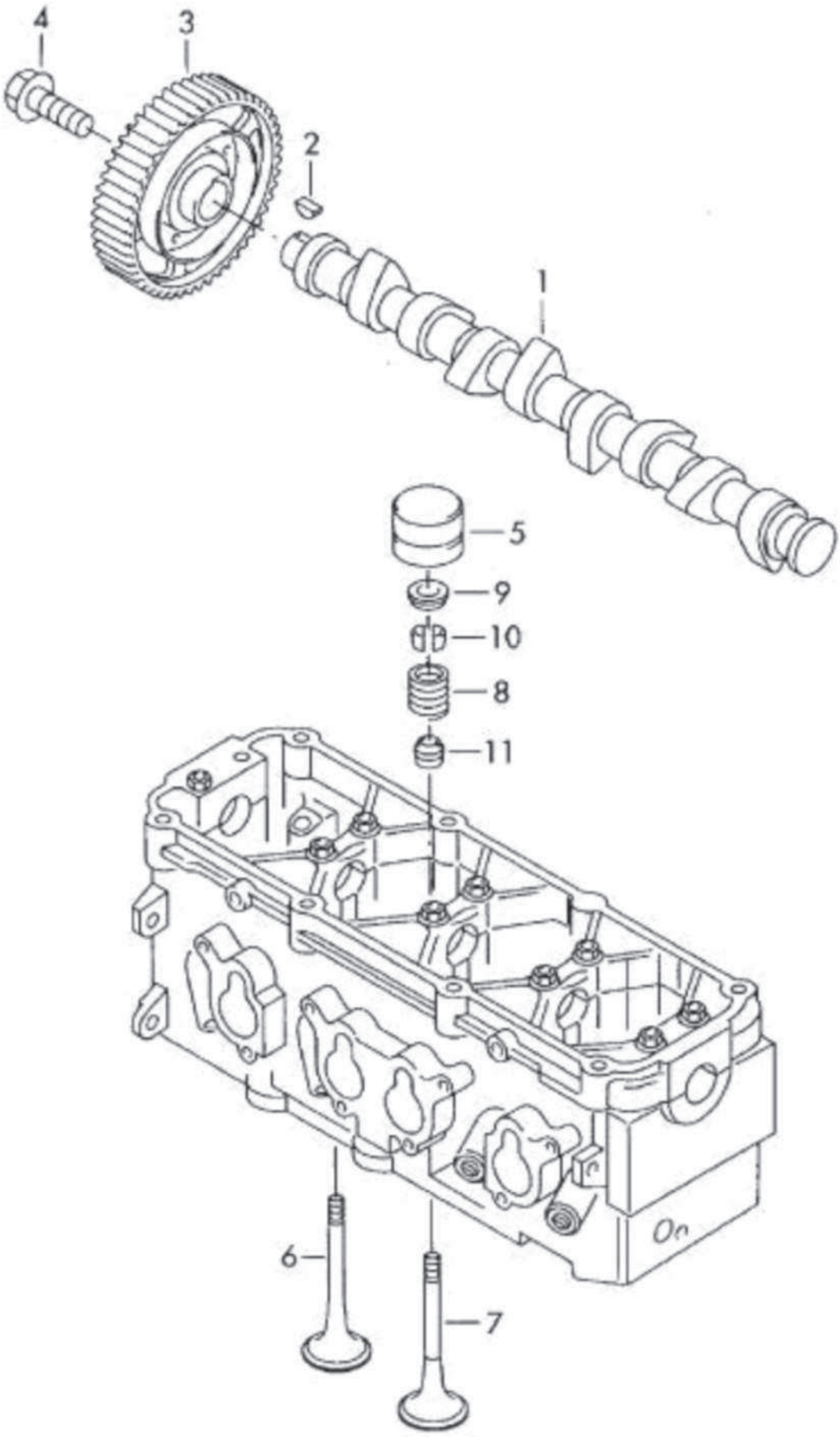
1.4. Cylinder head



cylinder head

POS	PART NUMBER	DESCRIPTION	QTY
1	06A 103 351 P	cylinder head	1
2	N 044 411 5	threaded pin M8X30	2
3	038 103 085 C	shaft oil seal (shaft rotation: to right)	1
4	044 103 499 B	pipe union	1
5	N 011 558 13	washer 8.4X15X1.6	10
6	N 011 008 46	hexagon nut M8	10
7	N 044 411 5	threaded pin M8X30	8
8	035 103 379	bungs	1
9	06G 103 383 A	cylinder head gasket (metal)	1
10	06A 103 384 C	socket head bolt M10X1.5X115	10
11	N 011 914 10	end cap	1
12	056 103 397	threaded pin M8X56	10
13	N 044 462 5	threaded pin M6X25	8
14	037 103 419 B	valve guide 36.5MM/7MM	8
15	06A 103 475 C	cylinder head cover	1
16	06A 103 544	oil deflector	1
17	056 103 591	strip	1
18	050 103 591	strip	1
19	N 015 082 10	hexagon collar nut M6	8
20	051 103 483 A	seal	1
21	06A 103 465	breather assy.	1
22	06A 103 483 D	seal	1
23	06B 103 485 C	end cap	1
24	06A 103 483 D	seal	1
25	06A 905 161 B	hall sensor	1
26	N 014 701 11	hexagon socket head bolt M6X12	1

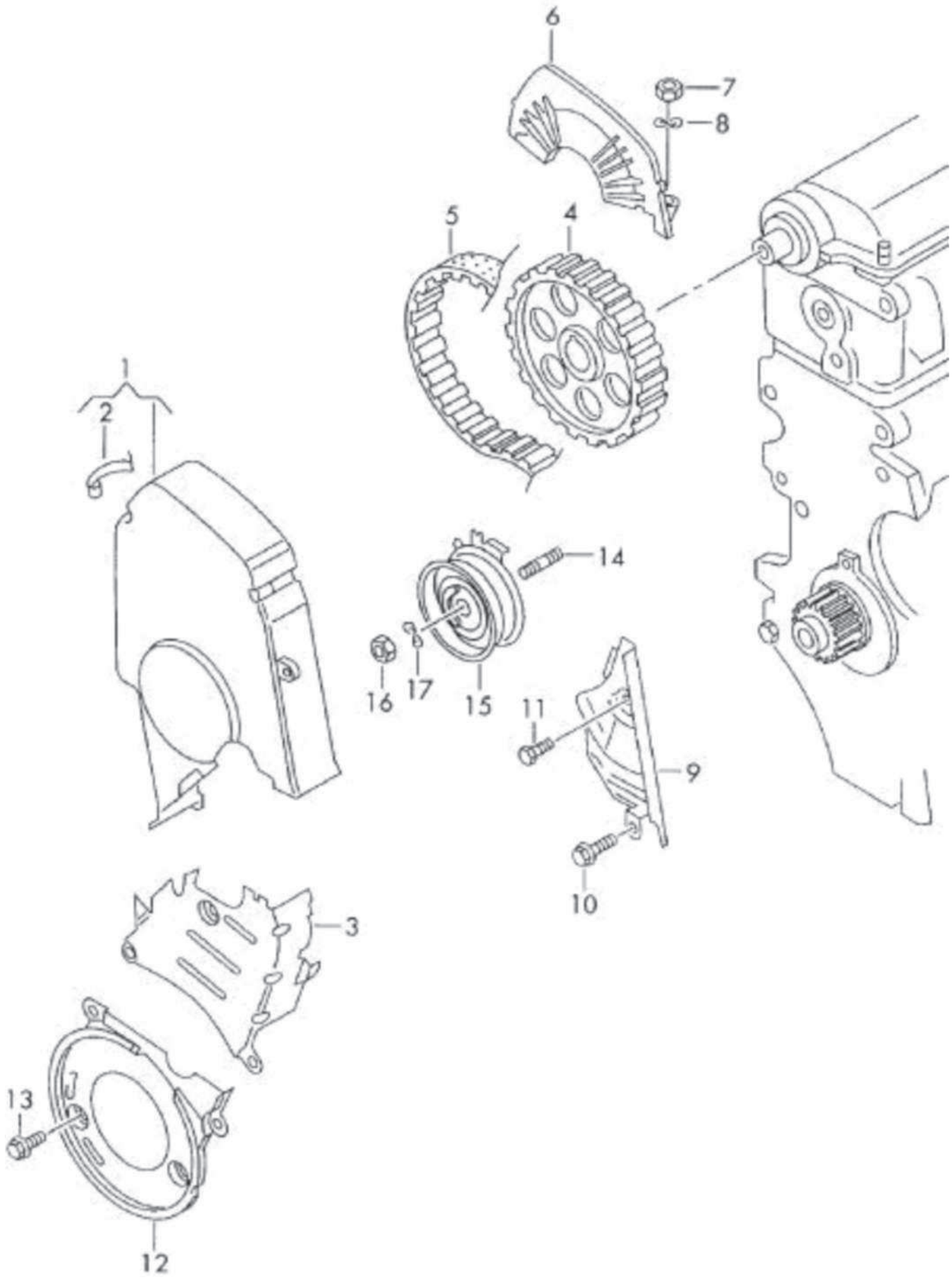
1.5. Camshaft / valves



camshaft / valves

POS	PART NUMBER	DESCRIPTION	QTY
1	050 109 101 P	Camshaft original	1
1	EST P/N 2 05 900	Camshaft BHKW	1
2	N 012 708 2	woodruff key	1
3	06A 109 105 C	toothed belt pulley	1
4	N 906 340 01	hexagon head bolt M12X1.5X30	1
5	050 109 309 H	tappets	8
6	06A 109 601 E	inlet valve	4
7	048 109 611 B	exhaust valve	4
8	078 109 623 C	valve spring	8
9	078 109 641 C	spring disc	8
10	021 109 651	cotter / split	16
11	027 109 675	valve stem seal	8

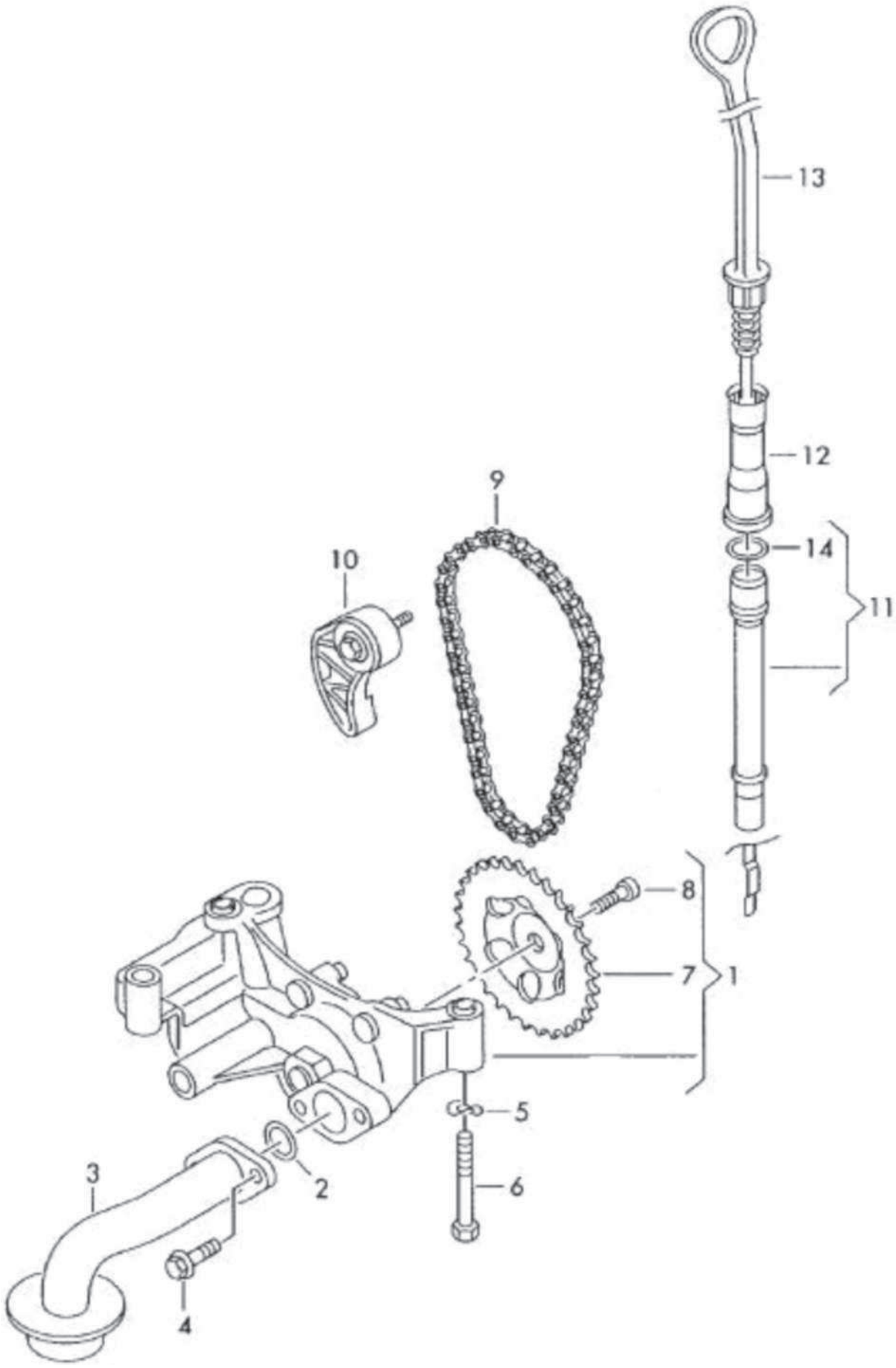
1.6. Toothed belt / toothed belt guard



toothed belt / toothed belt guard

POS	PART NUMBER	DESCRIPTION	QTY
1	06A 109 108 T	toothed belt guard upper	1
2	06A 109 150	spring clip	2
3	06A 109 147 F	toothed belt guard	1
4	06A 109 105 C	toothed belt pulley	1
5	06A 109 119 C	toothed belt Z=138	1
6	048 109 173 D	cover upper	1
7	N 011 006 36	hexagon nut M6	2
8	N 011 524 26	washer 6.4X12.5X1.6	2
9	06A 109 129 A	cover	1
10	N 909 450 02	hexagon socket head bolt with shoulder M7X21.5	1
11	N 010 293 16	hexagon bolt M8X16	1
12	06A 109 175 B	toothed belt guard lower	1
13	N 101 563 05	hex collared bolt M6X12	5
14	N 014 408 3	threaded pin M10XM8X62	1
15	06A 109 479 F	tensioning roller	1
16	N 011 008 40	hexagon nut M8	1
17	N 012 241 10	spring washer B8X15	1
-	06A 198 119	Repair kit for toothed belt. Kit contains part#; 06A 109 119 C 06A 109 479 F N 901 076 03 N 011 008 40	1 1 1 1 1

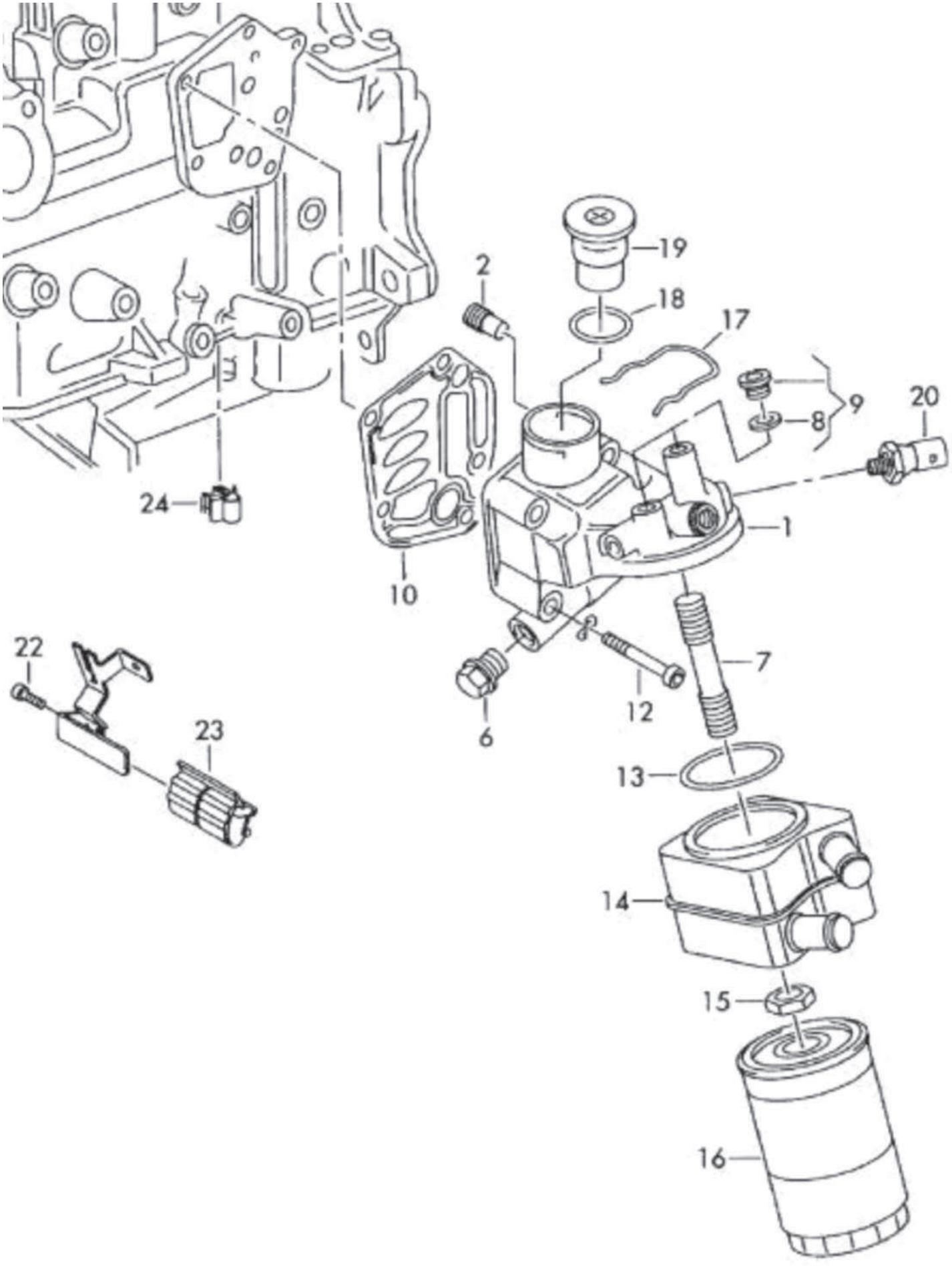
1.7. Oil pump / oil dipstick



oil pump / oil dipstick

POS	PART NUMBER	DESCRIPTION	QTY
1	06A 115 105 B	oil pump	1
2	N 028 222 2	o-ring 20X3.15	1
3	038 115 251 B	suction line	1
4	N 909 450 02	hexagon socket head bolt with shoulder M7X21.5	2
5	N 012 234 5	spring washer	3
6	N 907 040 07	hexagon socket head bolt with shoulder M7X55	3
7	06A 115 121	chain sprocket	1
8	N 903 663 03	cylinder screw self-locking M8X22	1
9	06A 115 125	roller chain	1
10	06A 115 130	chain tensioner	1
11	06D 103 634 E	tube for oil dipstick	1
12	06A 103 663 C	guide funnel	1
13	06B 115 611 R	oil dipstick	1

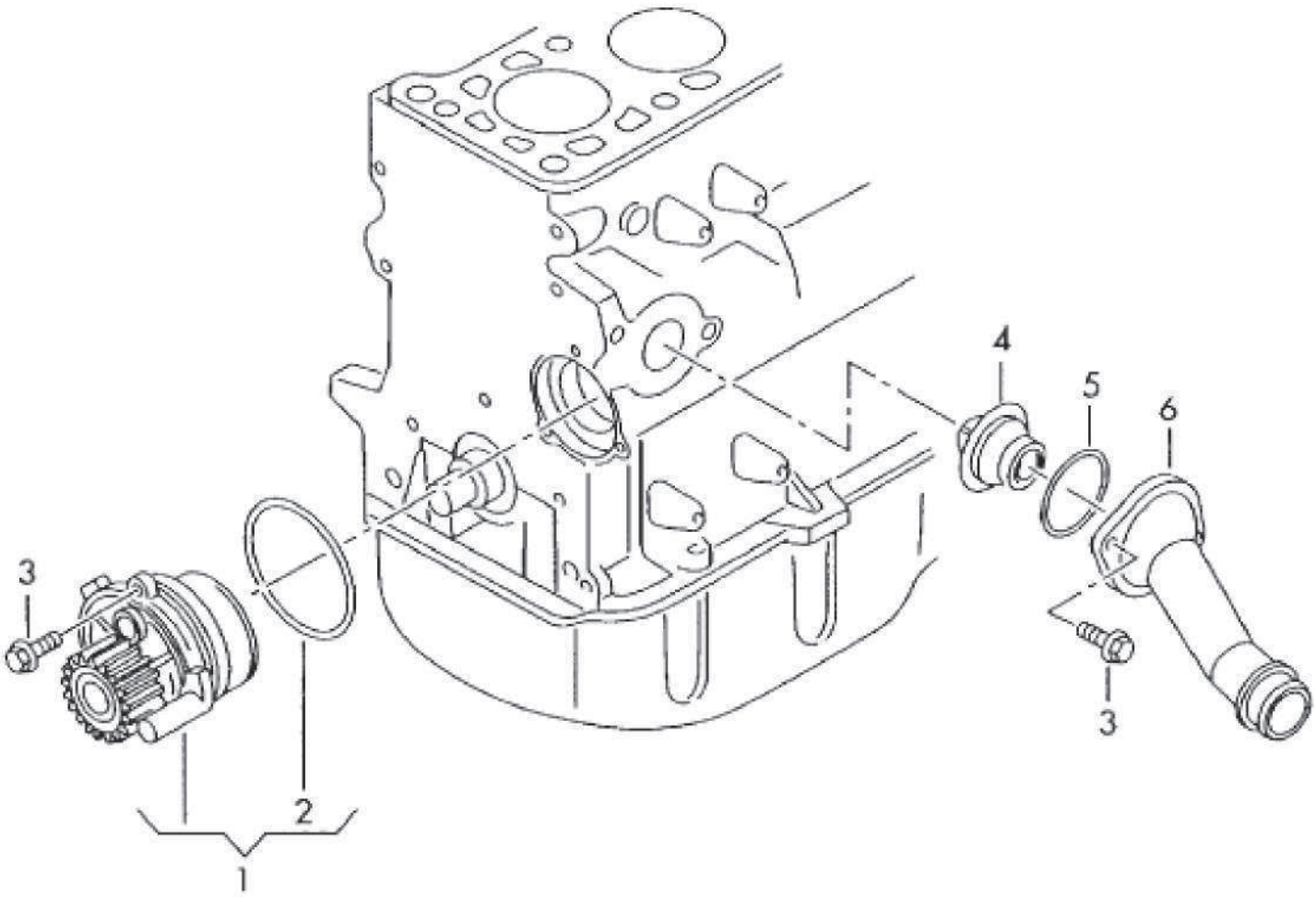
1.8. Oil filter bracket



oil filter bracket

POS	PART NUMBER	DESCRIPTION	QTY
1	06A 115 405 BP	oil filter bracket	1
2	06A 103 175	valve	1
6	WHT 000 422 A	sealing plug M20X1.5X10	1
7	028 115 721	threaded union	1
8	N 013 811 5	o-ring 10X13.5	2
9	N 016 024 3	sealing plug M10X1	2
10	06A 115 441 J	seal	1
12	N 907 040 07	hexagon socket head bolt with shoulder M7X55	4
13	038 117 070 A	o-ring 59X5	1
14	078 117 021 A	oil cooler	1
15	068 115 723	hexagon nut	1
16	06A 115 561 B	oil filter	1
17	037 121 142 A	retaining spring	1
18	N 904 673 01	o-ring 34.2X3	1
19	06A 115 418	end cap	1
20	038 919 081 M	oil pressure switch 0.75-1.05BAR	1
22	N 010 210 20	hexagon bolt M6X10	1
23	1H0 971 615 AB	cable holder	1
24	1H0 971 847 A	clip 23X46	1

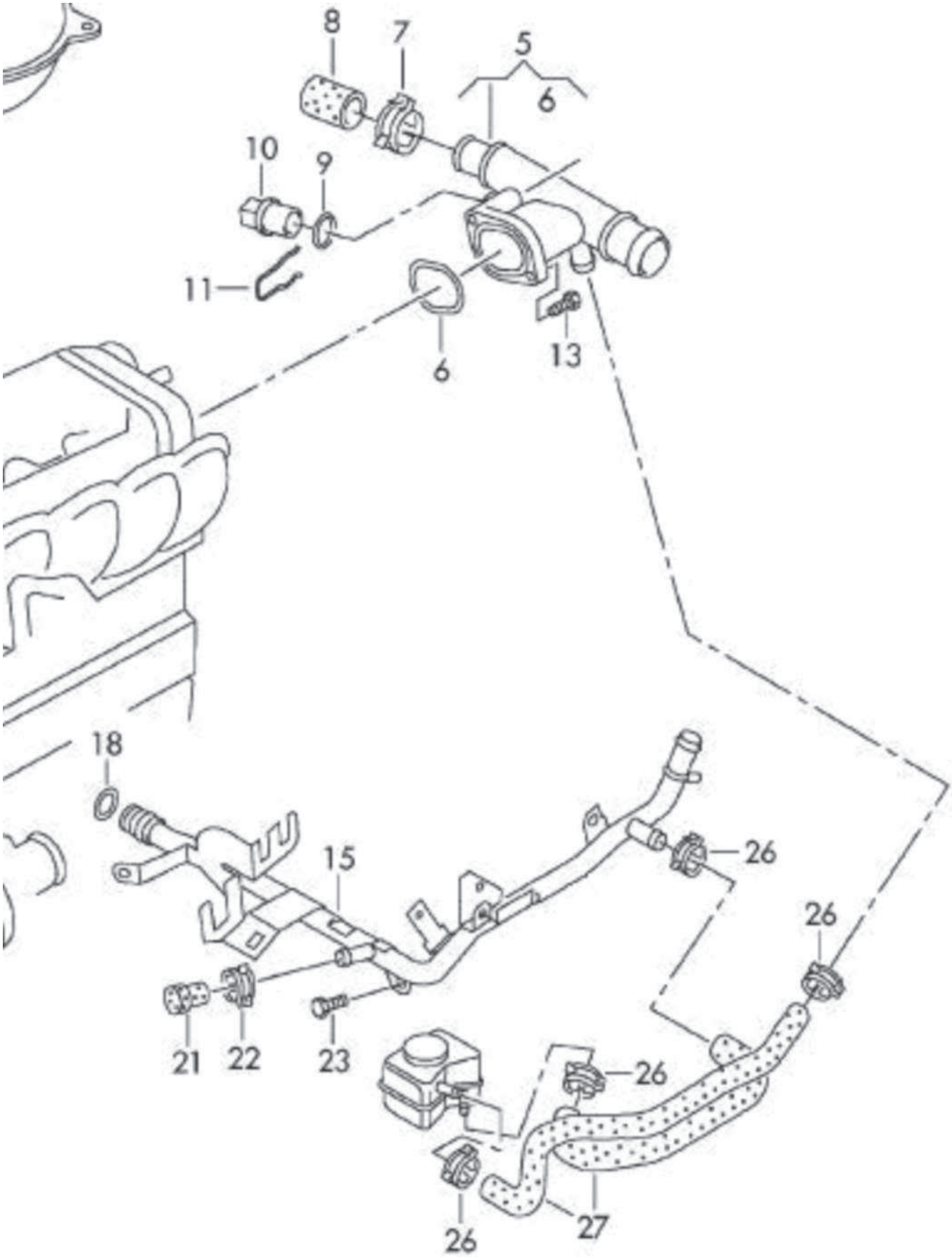
1.9. Coolant pump



coolant pump

POS	PART NUMBER	DESCRIPTION	QTY
1	06A 121 012 G	coolant pump with sealing ring original	1
2	06A 121 119	o-ring 53X4MM	1
3	N 909 450 02	hexagon socket head bolt with shoulder M7X21.5	5
4	050 121 113 C	coolant regulator 87C-102C	1
5	038 121 119 B	o-ring 50X4	1
6	06A 121 121 C	pipe union	1

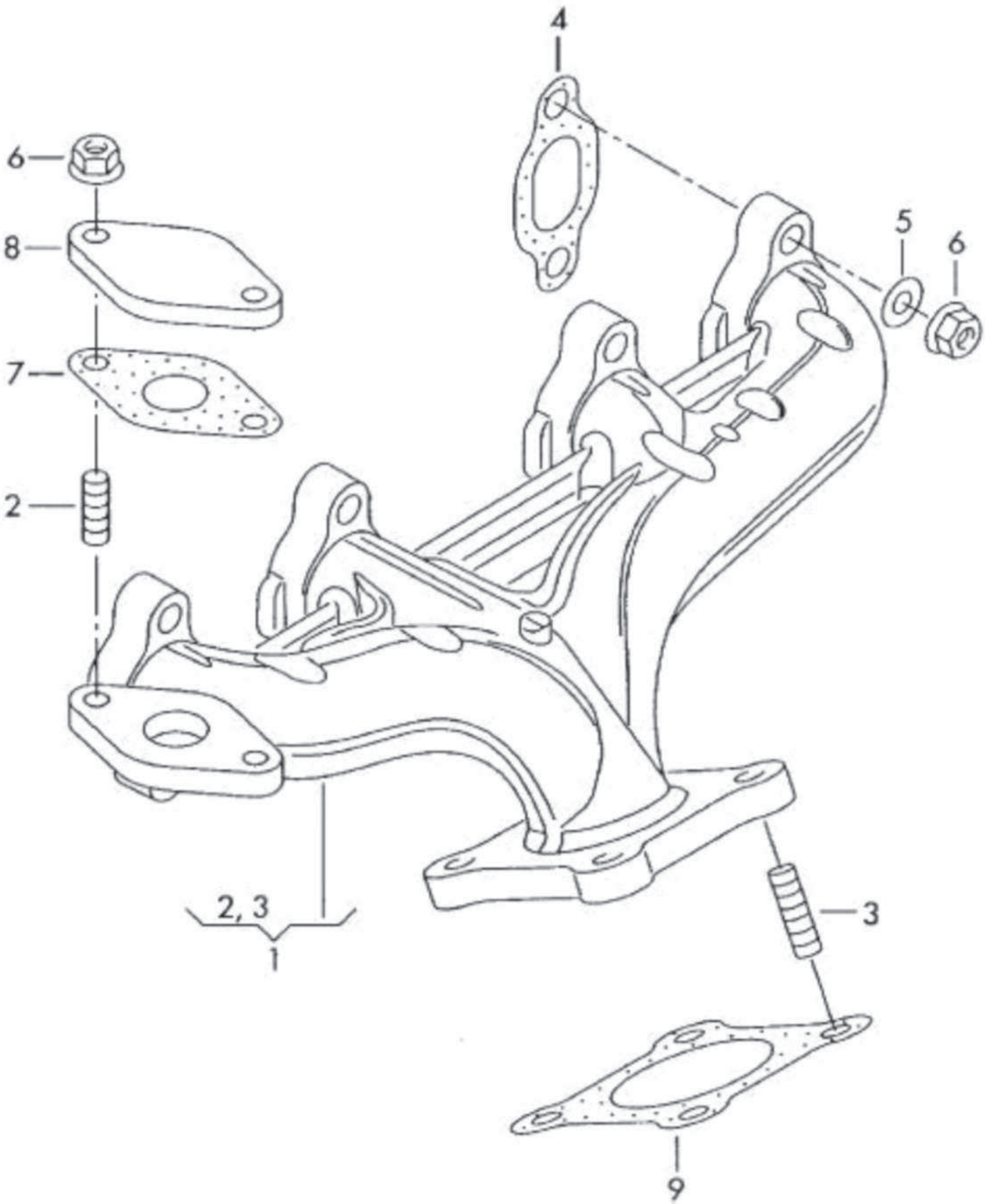
1.10. Coolant pipe



coolant pipe

POS	PART NUMBER	DESCRIPTION	QTY
5	06A 121 132	flange with sealing ring	1
6	037 121 688	o-ring 50X3.15	1
7	06B 145 917	spring band clamp 35X12	1
8	036 121 324 B	cap closure	1
9	N 903 168 02	o-ring 19.6X3.65	1
10	059 919 501 A	double temperature sensor 4 pin green	1
11	032 121 142	retaining spring	1
13	N 911 480 01	double stud with hexagon drive M6X22/M6X14	2
13	N 011 524 26	washer 6.4X12.5X1.6	1
13	N 015 082 10	hexagon collar nut M6	1
15	06A 121 065 AR	coolant pipe	1
18	N 903 653 02	o-ring 20X3	1
21	1J0 121 324	cap closure	2
22	N 906 869 02	spring band clamp 23X12	1
23	N 901 752 05	hexagon collar nut M6	2
26	N 906 869 02	spring band clamp 23X12	4
27	1C0 121 086 L	coolant hose	1

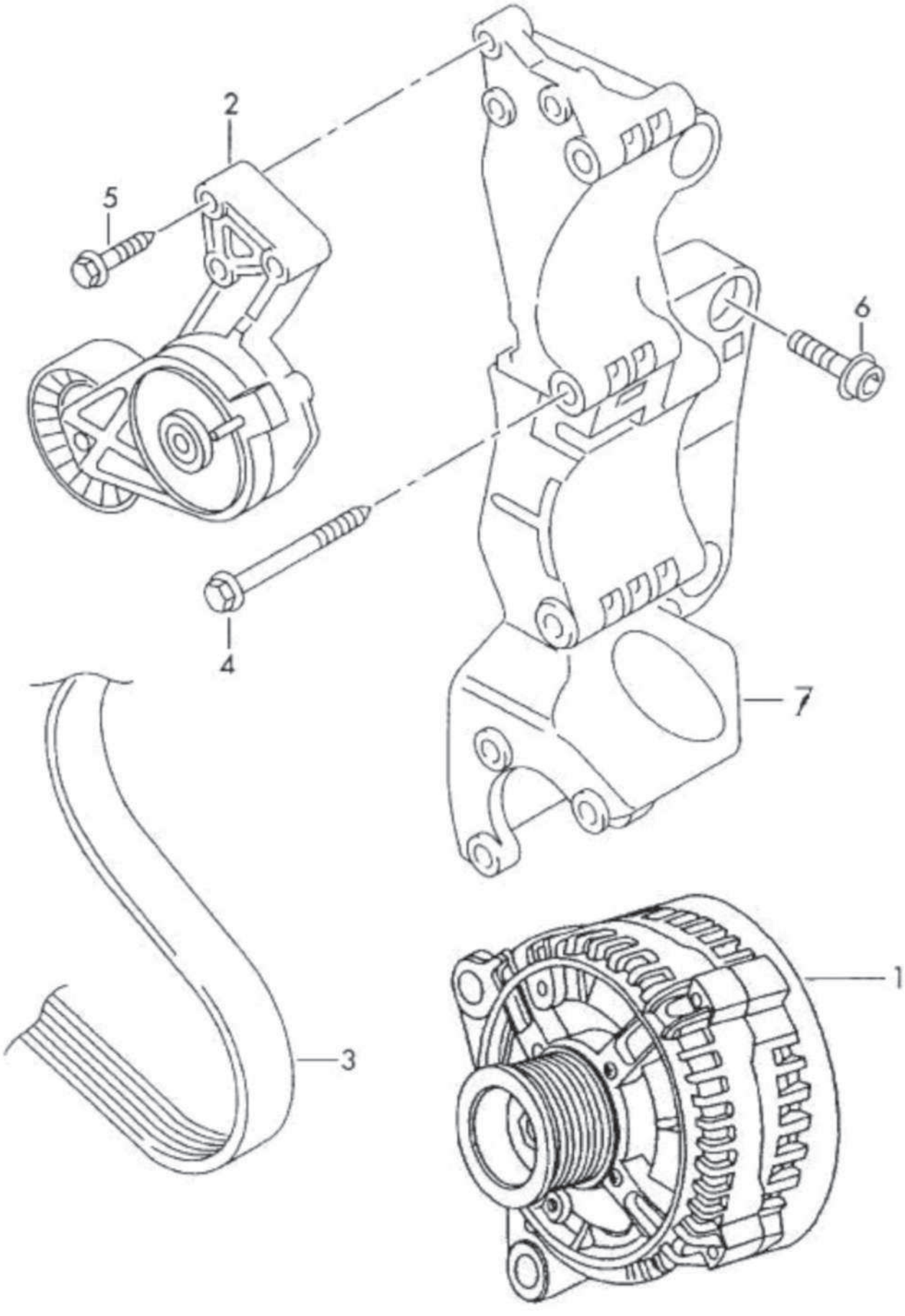
1.11. Exhaust manifold



exhaust manifold

POS	PART NUMBER	DESCRIPTION	QTY
1	028 253 031 AM	exhaust manifold	1
2	N 044 514 5	threaded pin M8X25	2
3	N 044 520 3	threaded pin M10X28	4
4	037 253 039 F	seal	1
5	N 900 955 01	washer 8.4X15X4	8
6	N 911 308 01	hexagon collar nut self-locking M8	10
7	028 131 547 G	seal	1
8	038 131 925 A	end cap	1
9	155 253 115	seal	1

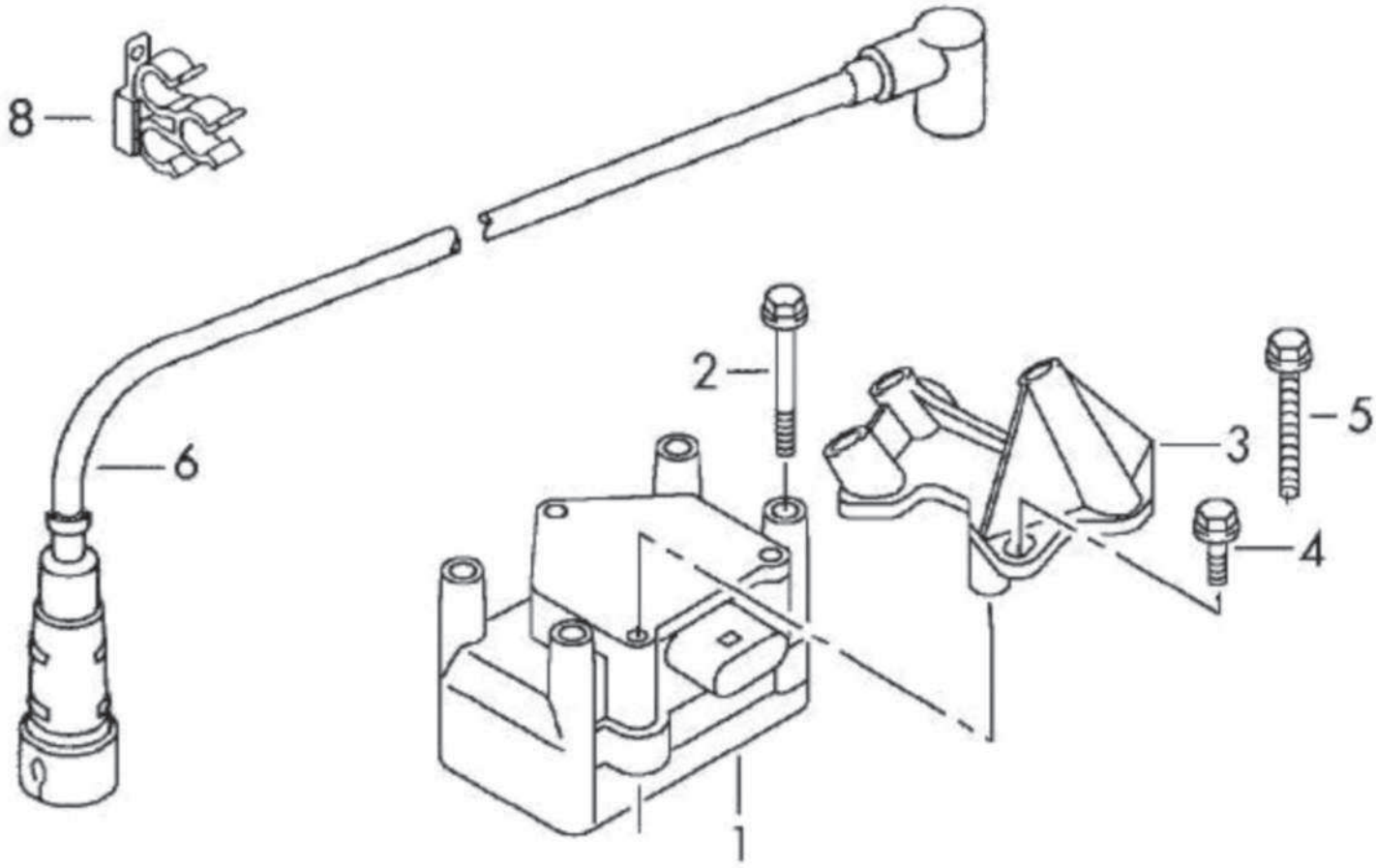
1.12. Alternator + mounting parts



alternator + mounting parts

POS	PART NUMBER	DESCRIPTION	QTY
1	028 903 028 C	alternator 14V/70A	1
1	028 903 028 D	alternator 14V/90A	1
1	03L 903 023 G	alternator 14V/110A	1
1	06F 903 023 F	alternator 14V/140A	1
2	06A 903 315 E	tensioner	1
3	06A 145 933 L	poly-v-belt 21.36X926MM	1
4	N 106 100 01	hex collared bolt M8X90	2
5	N 019 543 3	hex collared bolt M8X50	3
6	N 907 529 02	socket head bolt with inner multipoint head M10X45	6
7	06A 903 141 P	bracket for alternator and refrigerant compressor	1

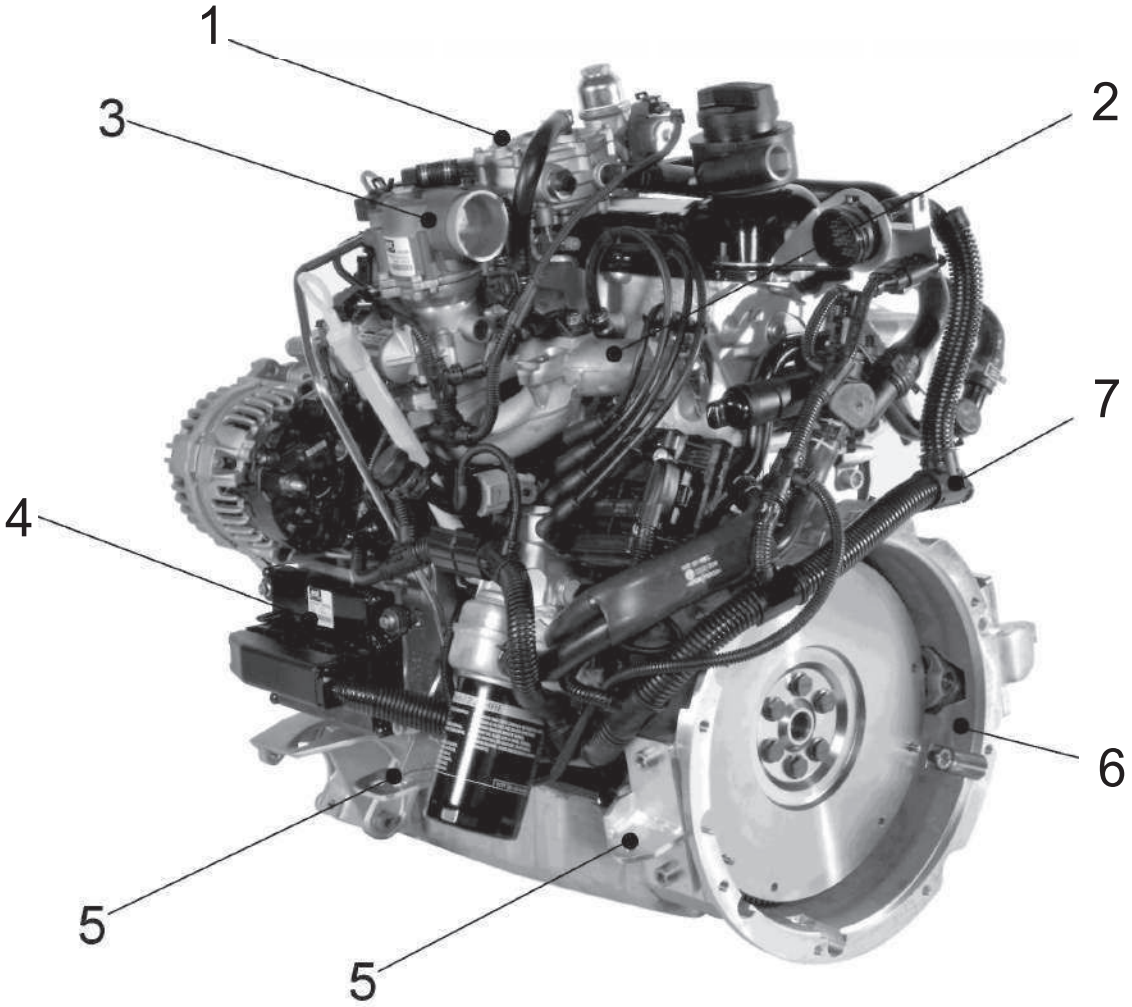
1.13. Ignition

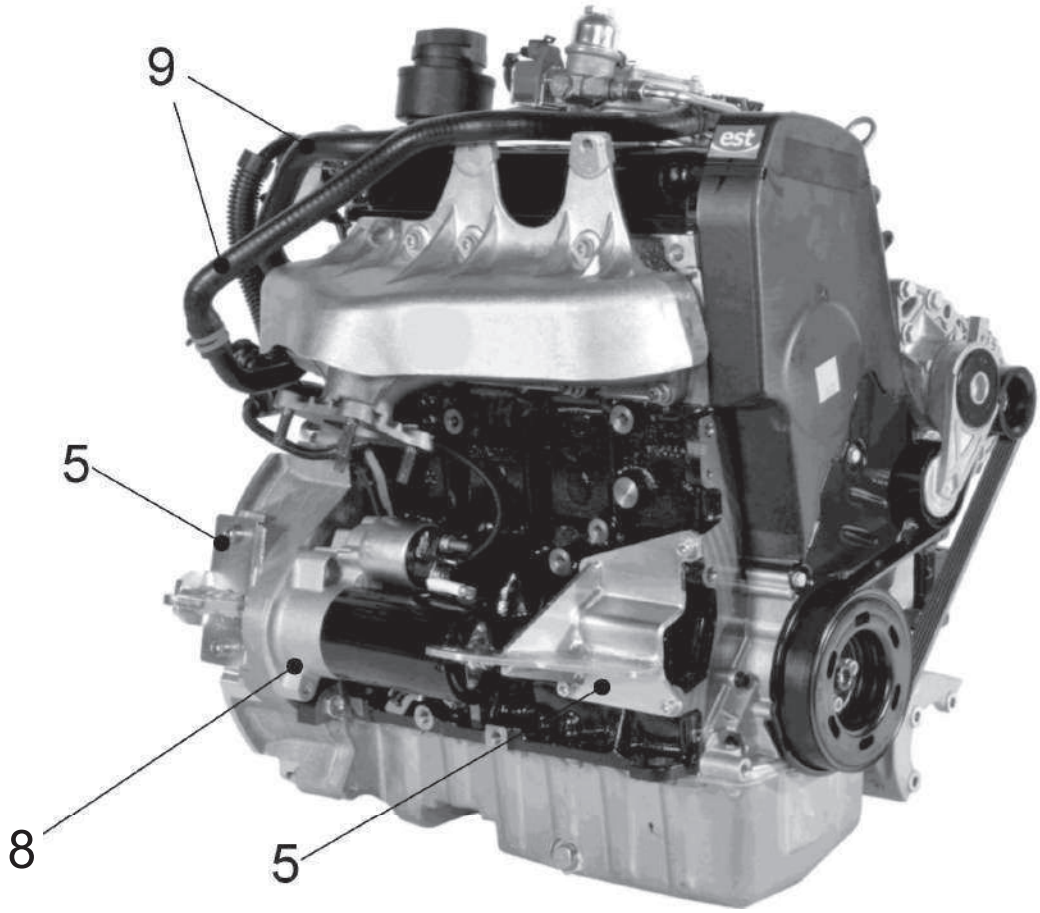


ignition

POS	PART NUMBER	DESCRIPTION	QTY
1	06A 905 097	ignition coil !!!**This part is no longer available!!! Contact your supplier**	1
2	N 105 626 03	round hd. screw M6X60	3
3	06A 903 103 B	bracket	1
4	N 909 450 02	hexagon socket head bolt with shoulder M7X21.5	2
5	N 907 040 07	hexagon socket head bolt with shoulder M7X55	1
6	06A 905 430 AK	ignition lead cylinder 1	1
6	06A 905 430 AL	ignition lead cylinder 2	1
6	06A 905 430 AM	ignition lead cylinder 3	1
6	06A 905 430 AN	ignition lead cylinder 4	1
6	06A 905 409 N	1 set: ignition leads cylinders 1-4	1
8	050 905 471 A	retainer for ignition cables	3

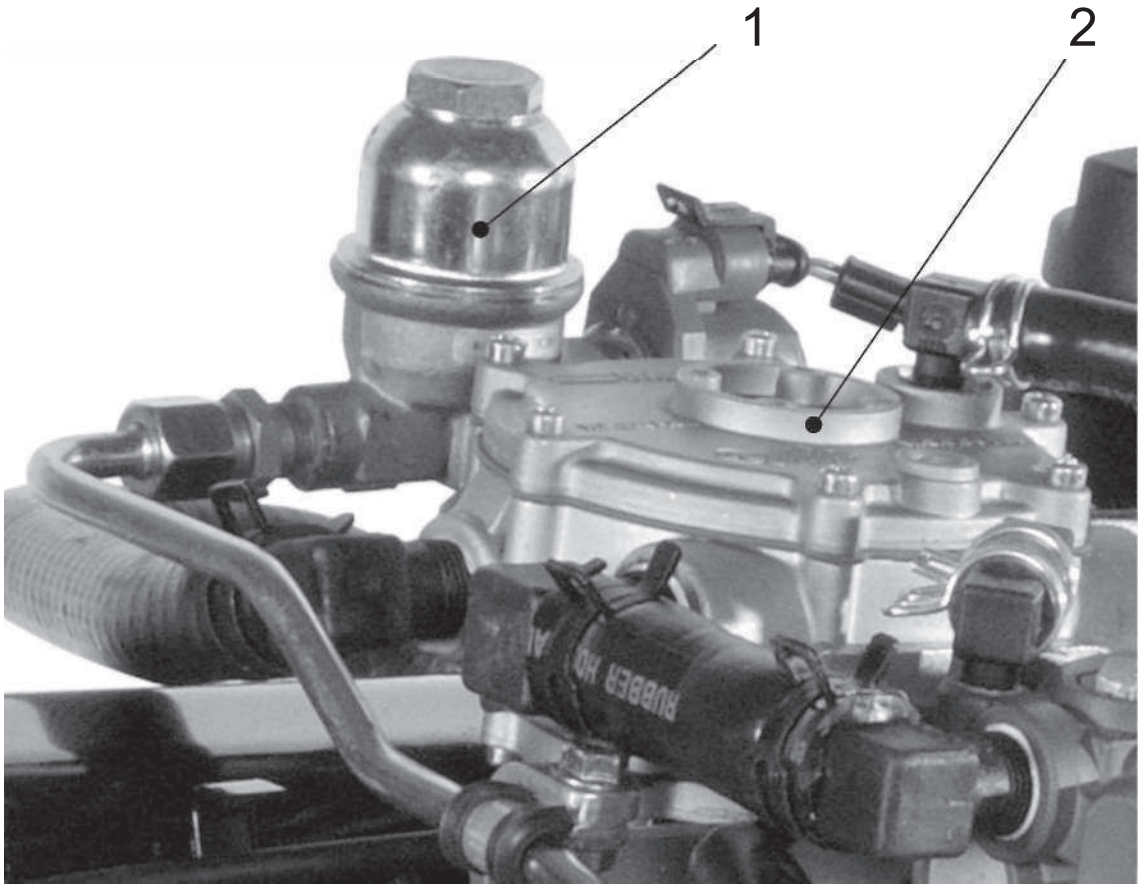
2. SPECIFIC PARTS

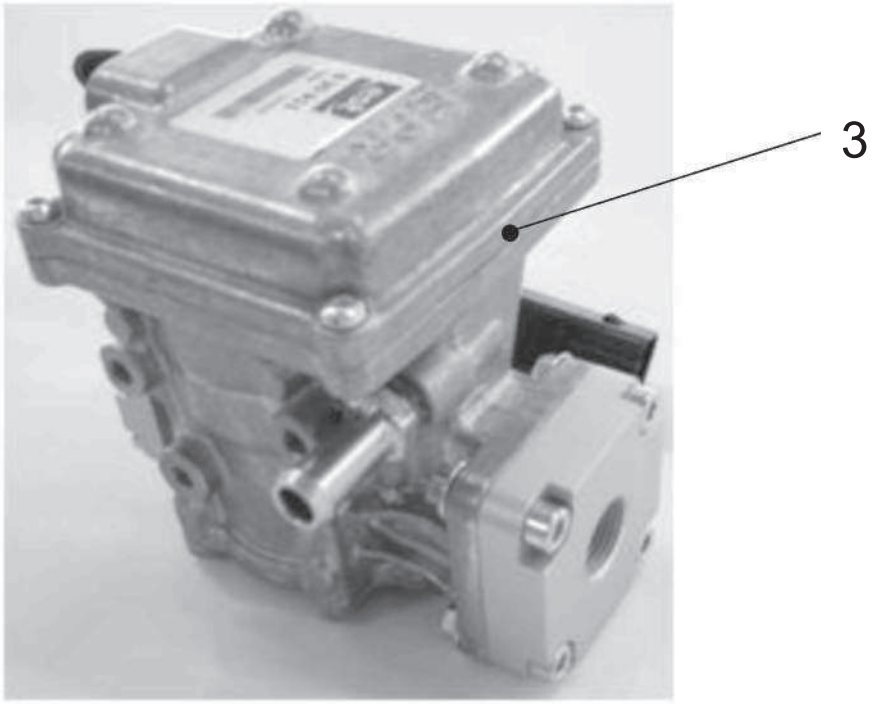




Specific parts	
POS	**DESCRIPTION**
1	Pressure regulator
2	Intake module
3	Gas mixer
4	ECU
5	Engine brackets
6	Flywheel housing
7	Harness
8	Starter
9	Water hoses
10	Gasoline fuel parts

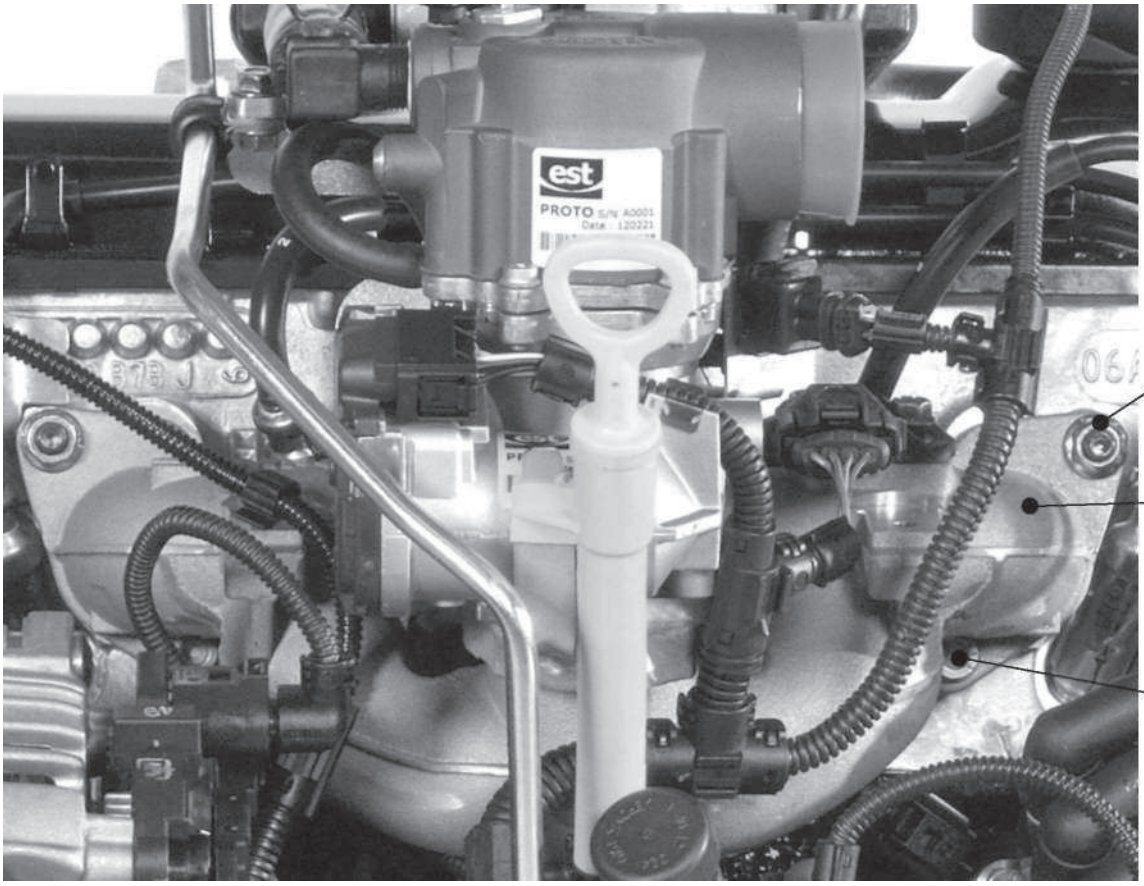
2.1. Pressure regulator + mounting parts

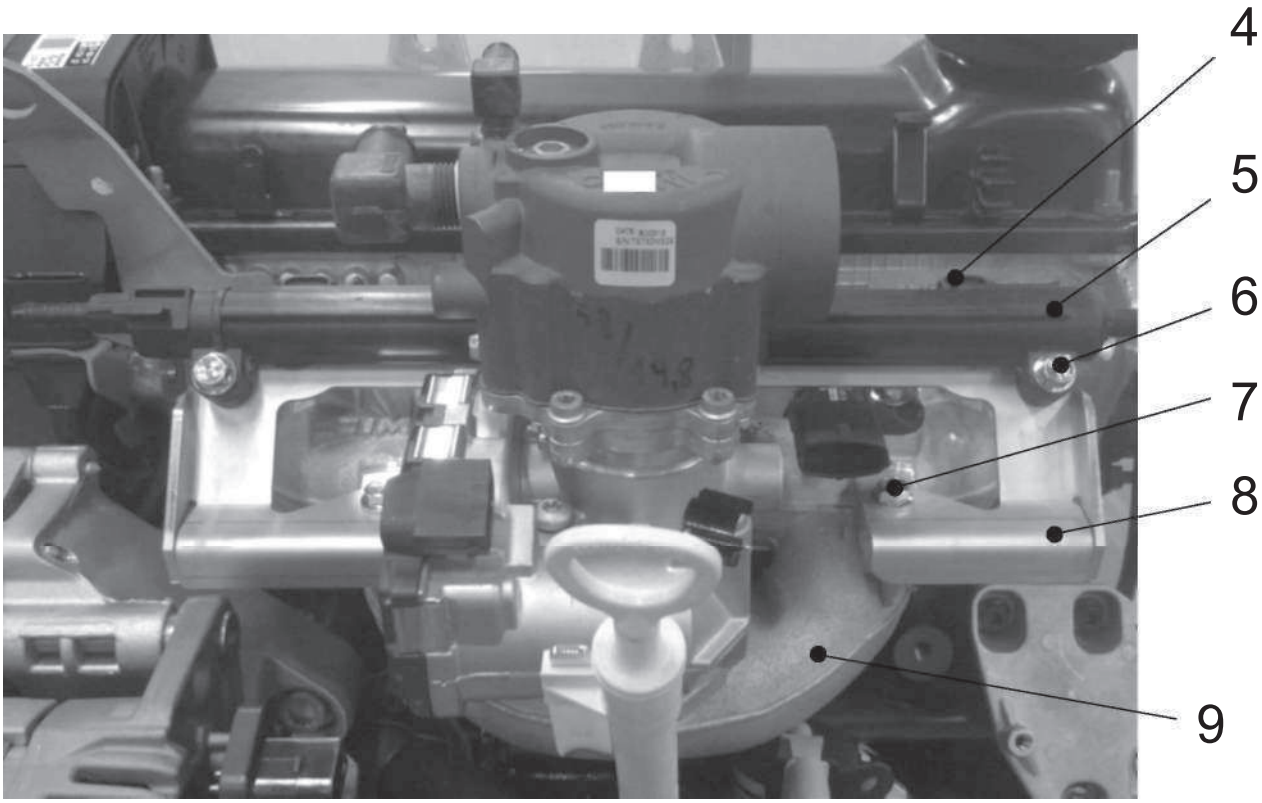




pressure regulator + mounting parts			
POS	**PART NUMBER**	**DESCRIPTION**	**QTY**
1	6 40 900	Lock off valve	1
2	4 10 900	Pressure regulator	1
3	6 30 910	Electronic pressure regulator	1
3	6 30 911	Electronic pressure regulator + adapters	1
-	4 00 900	Kit, pressure regulator, coolant hoses and mounting parts.	1

2.2. Intake module + mounting parts

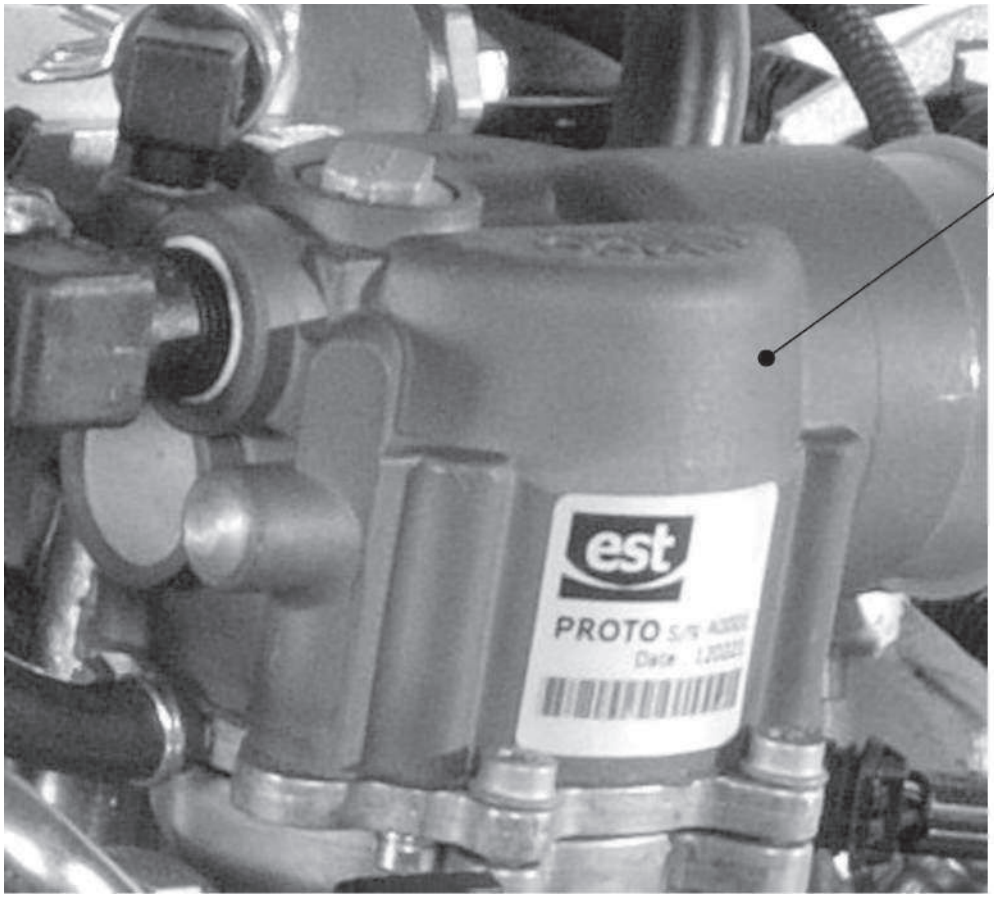


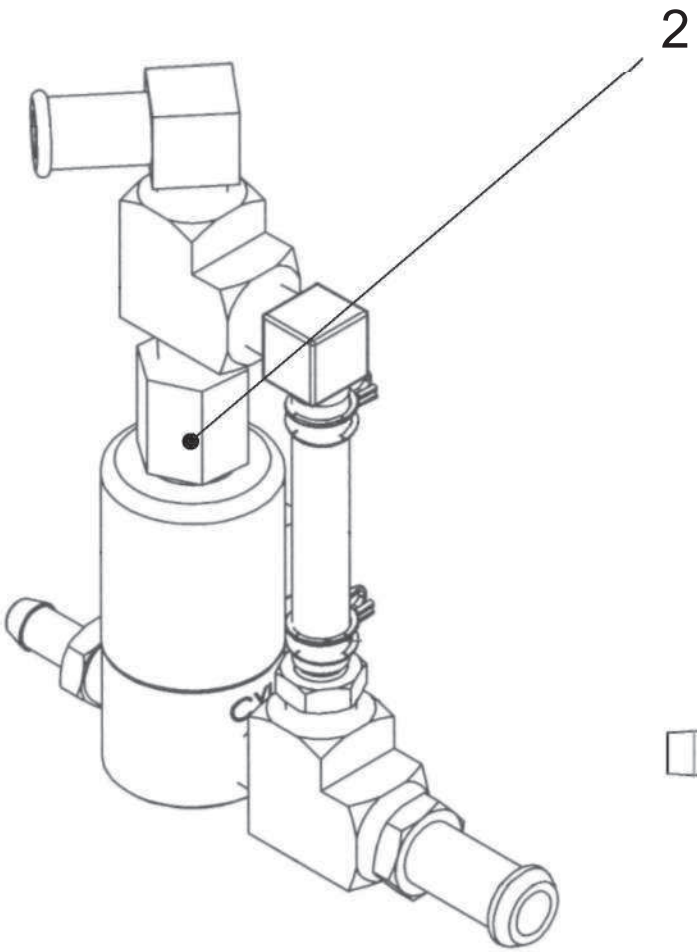


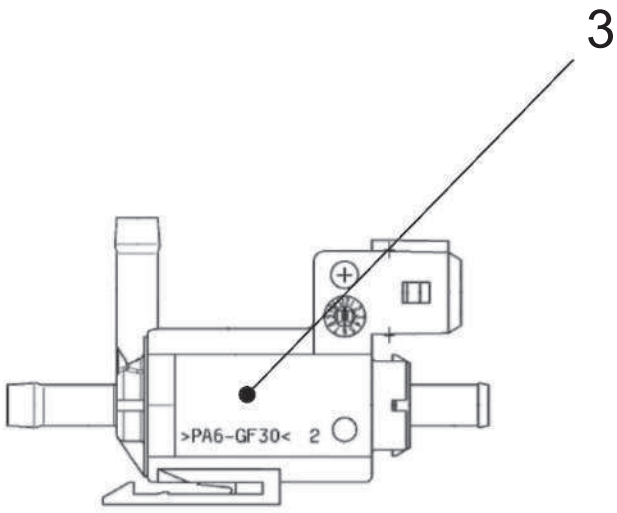
intake module LPG + mounting parts			
POS	**PART NUMBER**	**DESCRIPTION**	**QTY**
1	See kit below	Nut, flange M8	2
2	See kit below	Bolt, sockethead flangebolt M8x25 intake	6
3	Customer specific, *contact your supplier for further information*	Intake module LPG complete with manifold, throttle body, gas mixer, map sensor, gaskets and mounting parts.	1
-	VW P/N 037 129 717 C	Gasket, intake manifold BEF/CBS	1

intake module gasoline + mounting parts			
POS	**PART NUMBER**	**DESCRIPTION**	**QTY**
4	VW P/N 06A 906 031 CE	injector	4
	VW P/N 037 133 047	Clamp, spring injector mounting	4
5	VW P/N 06A 133 317 AN	Fuel rail	1
	VW P/N 06G 133 335	Cap for fuel rail	1
6	See kit below	Bolt, sockethead M6x35	2
	See kit below	Nut, flange M6	2
7	See kit below	Bolt, sockethead M6x14	2
8	See kit below	Bracket fuel rail	1
	VW P/N 037 129 717 C	Gasket, intake manifold BEF/CBS	1
9	Customer specific, *contact your supplier for further information*	Intake module gasoline, complete with manifold, throttle body, gas mixer, map sensor, injectors, fuel rail, bracket, gaskets and mounting parts.	1

2.3. Gas mixer + fuel control valve

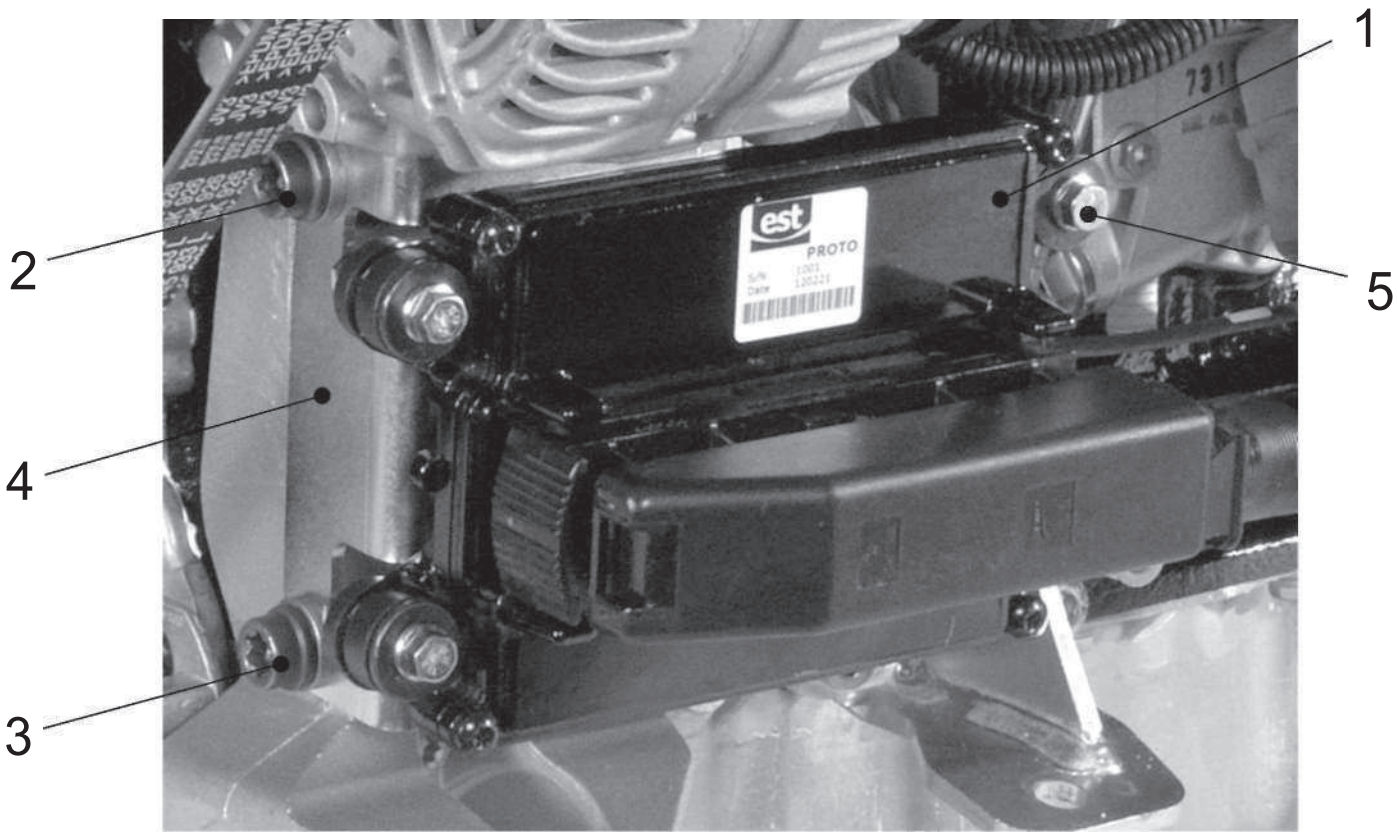


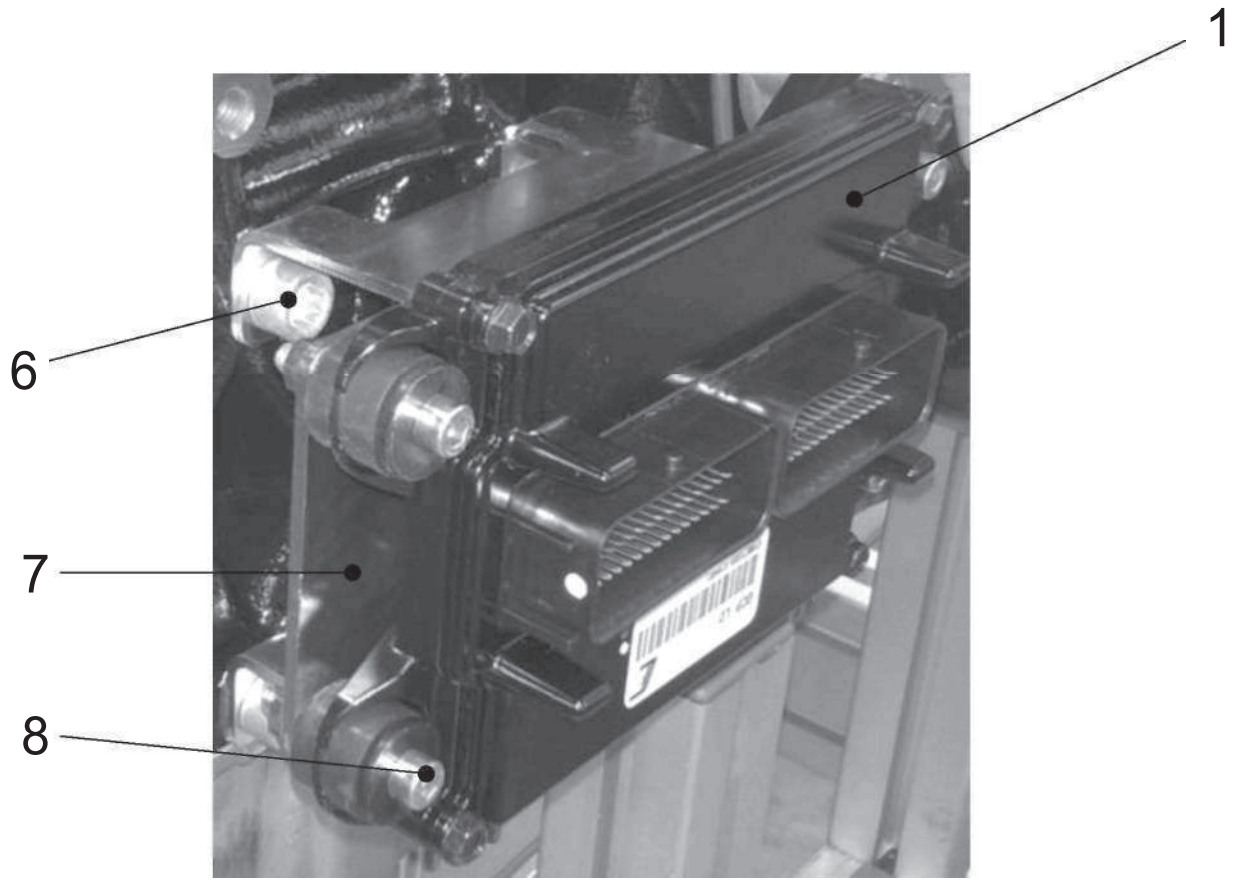




gas mixer + fuel control valve			
POS	**PART NUMBER**	**DESCRIPTION**	**QTY**
1	Customer specific, *contact your supplier*	Gas mixer	1
2	6 30 902	Fuel control valve	1
3	6 30 904	Fuel control valve	1

2.4. ECU + mounting parts

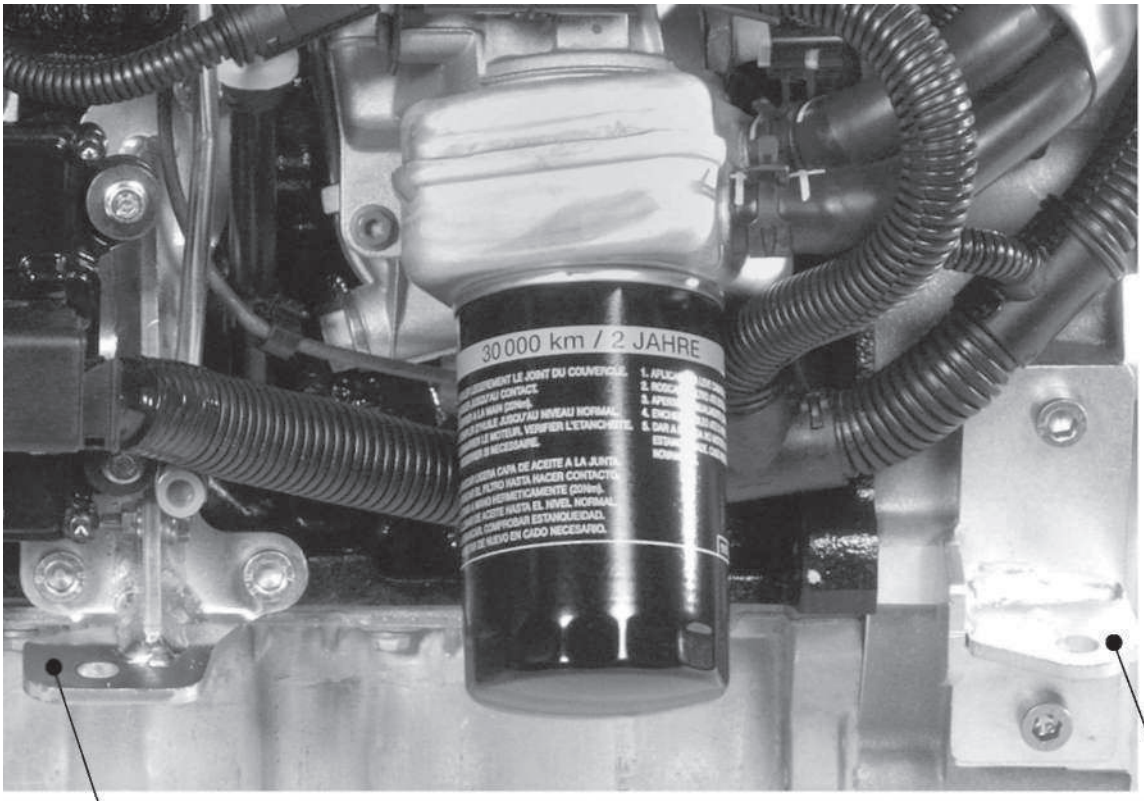




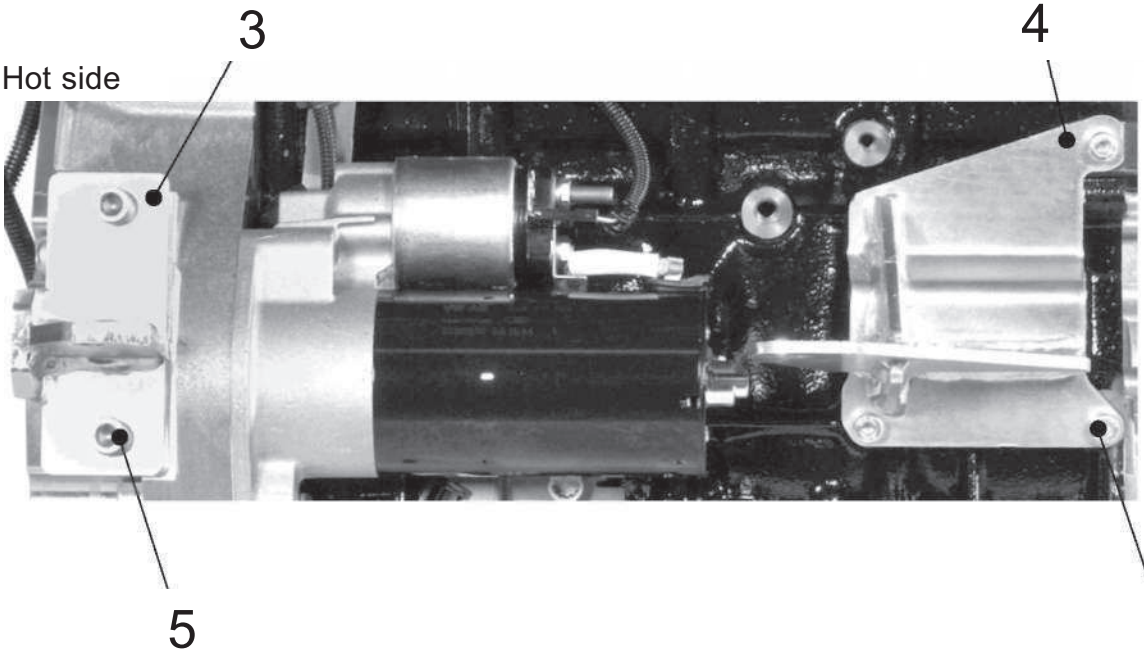
ECU + mounting parts			
POS	**PART NUMBER**	**DESCRIPTION**	**QTY**
1	Customer specific, *contact your supplier*	ECU, GCP90	1
2	See kit below	Bolt, sockethead M10x100 + Nut, M10	1
3	See kit below	Bolt, sockethead M10x60 + Nut, M10	1
4	See kit below	Bracket, ECU 2.0L airco position	1
5	See kit below	Bolt, sockethead M6x35 Nut, flange M6	4 4
-	9 95 900	Kit, mounting parts for ECU Kit contains: POS 2 + 3 + 4 + 5	1
6	See kit below	Bolt, multiple sockethead M10x12	4
7	See kit below	Bracket, ECU 2.0L bhkw	
8	See kit below	Bolt, sockethead M6x35 Nut, flange M6	4 4
-	9 95 901	Kit, mounting parts for ECU Kit contains: POS 6 + 7 + 8	

2.5. Engine brackets + mounting parts

Cold side



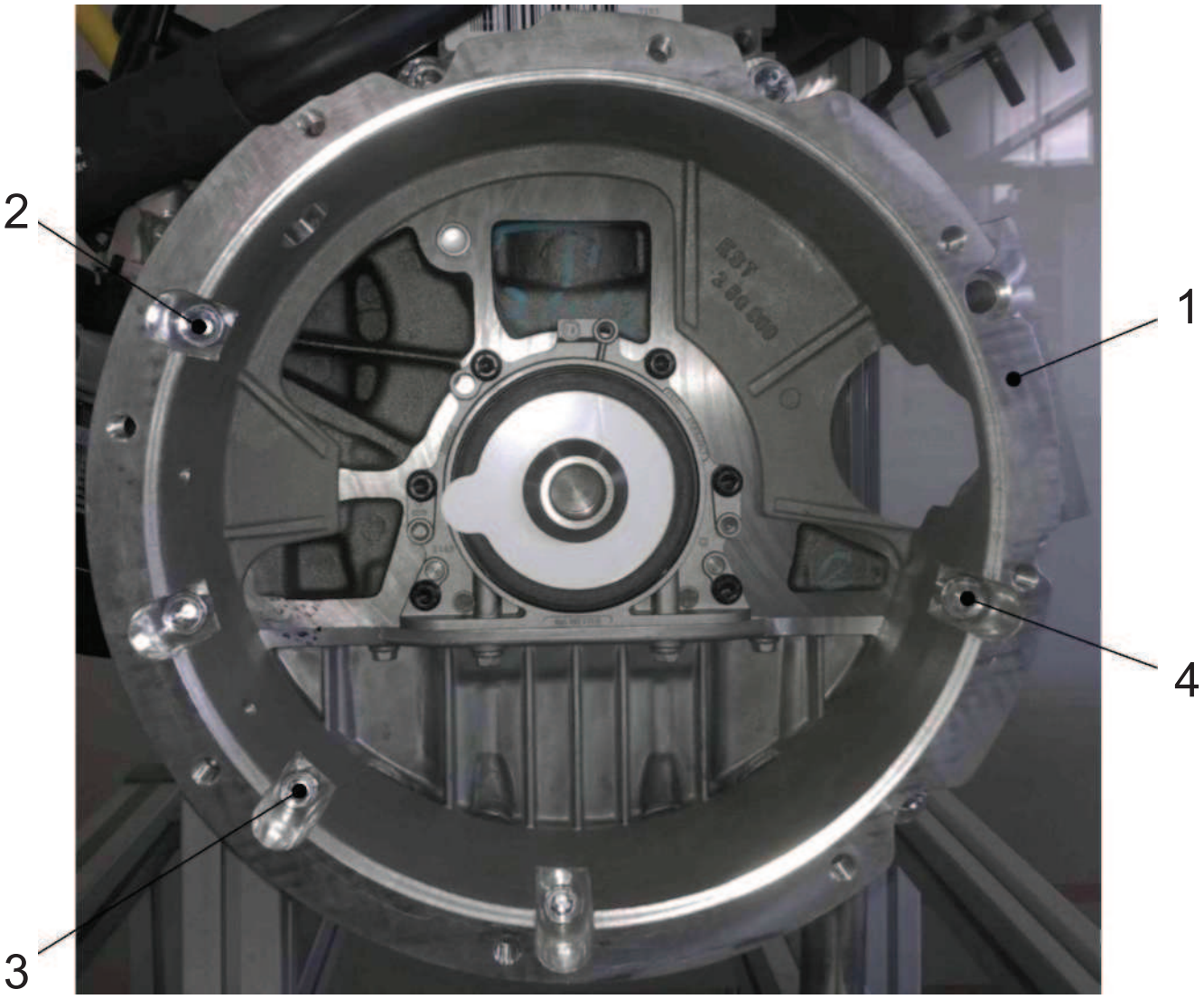
Hot side



engine brackets + mounting parts

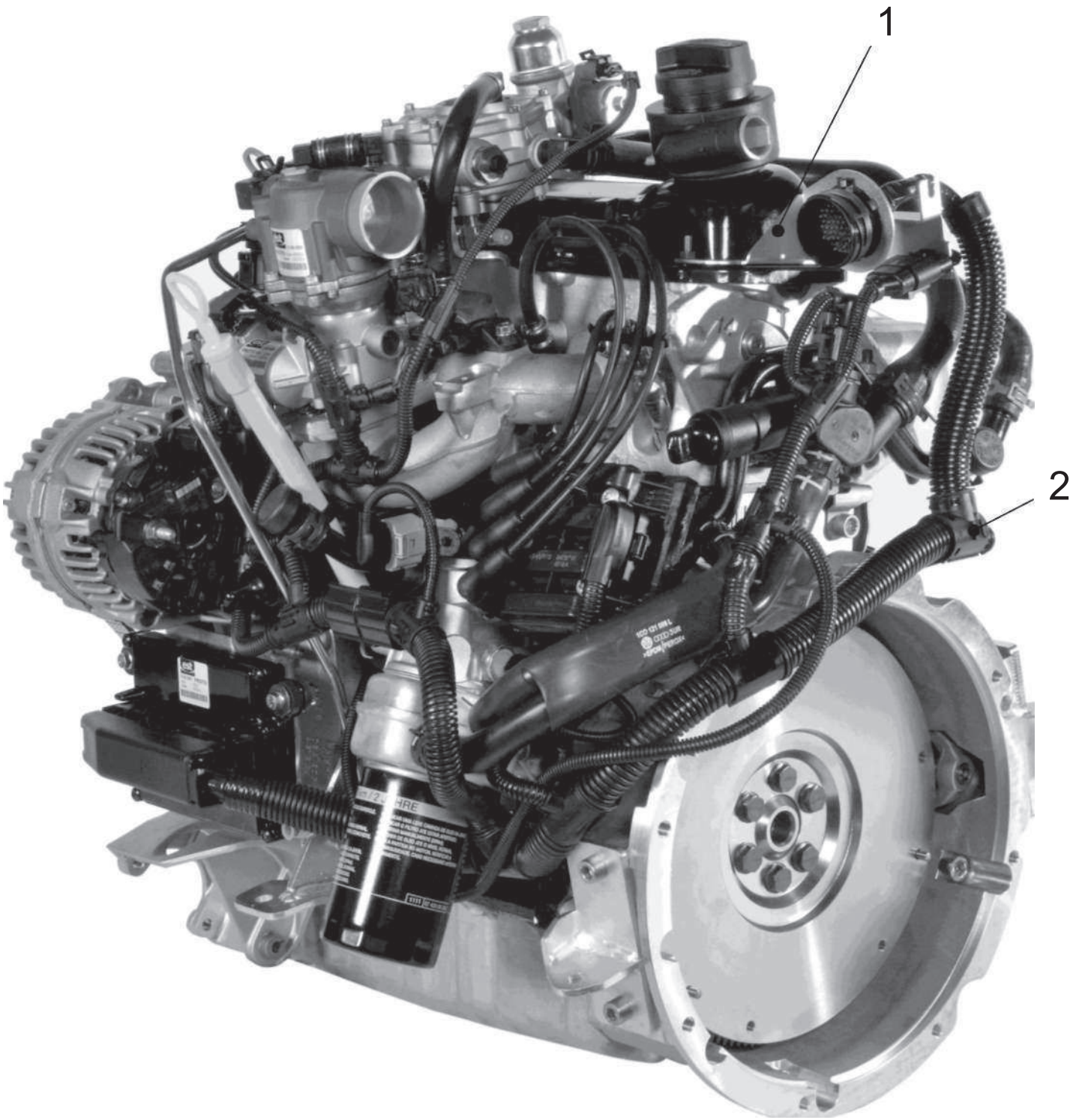
POS	PART NUMBER	DESCRIPTION	QTY
1	See kit below	Bracket, engine 2.0L intake side	1
2	See kit below	Bracket, engine 2.0L intake side flywhl	1
3	See kit below	Bracket, engine 2.0L exhaust side flywhl	1
4	See kit below	Bracket, engine 2.0L exhaust side	1
5	See kit below	Bolt, sockethead M8x20	3
6	See kit below	Bolt, sockethead M10x20	7
-	9 91 001	Kit, engine brackets with mounting parts Kit contains: POS 1 + 2 + 3 + 4 + 5 + 6	1

2.6. Flywheel housing + mounting parts



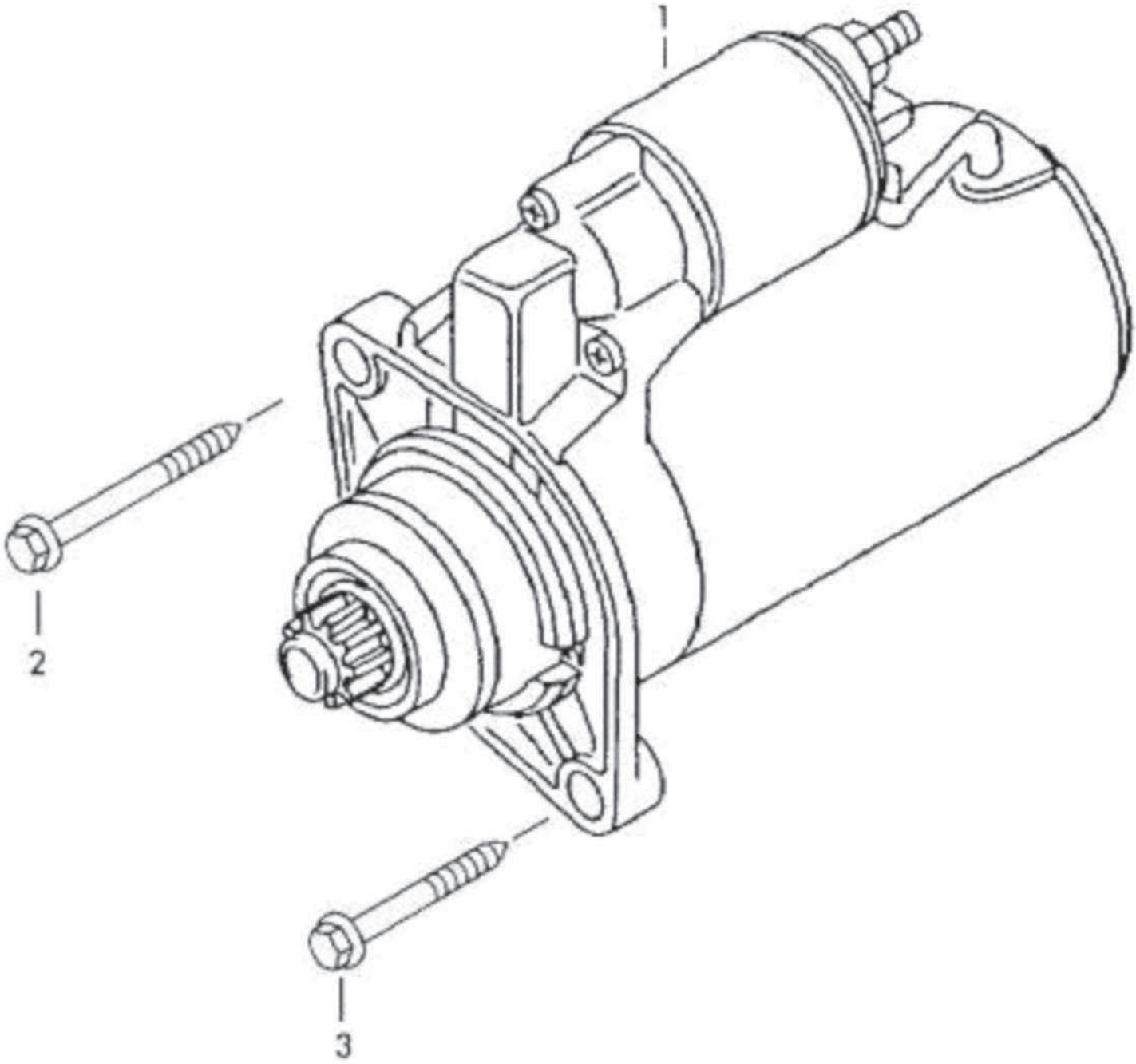
flywheel housing + mounting parts			
POS	**PART NUMBER**	**DESCRIPTION**	**QTY**
1	See kit below	Flywheel housing, EST SAE#5 BEF/CBS	1
2	See kit below	Bolt, sockethead M12x60	4
3	See kit below	Bolt, sockethead M10x60	3
4	See kit below	Bolt, sockethead M12x80	1
		Nut, M12 zinc plated	1
-	9 91 002	Kit, flywheel housing + mounting parts Kit contains: POS 1 + 2 + 3 + 4	1

2.7. Harness + mounting parts



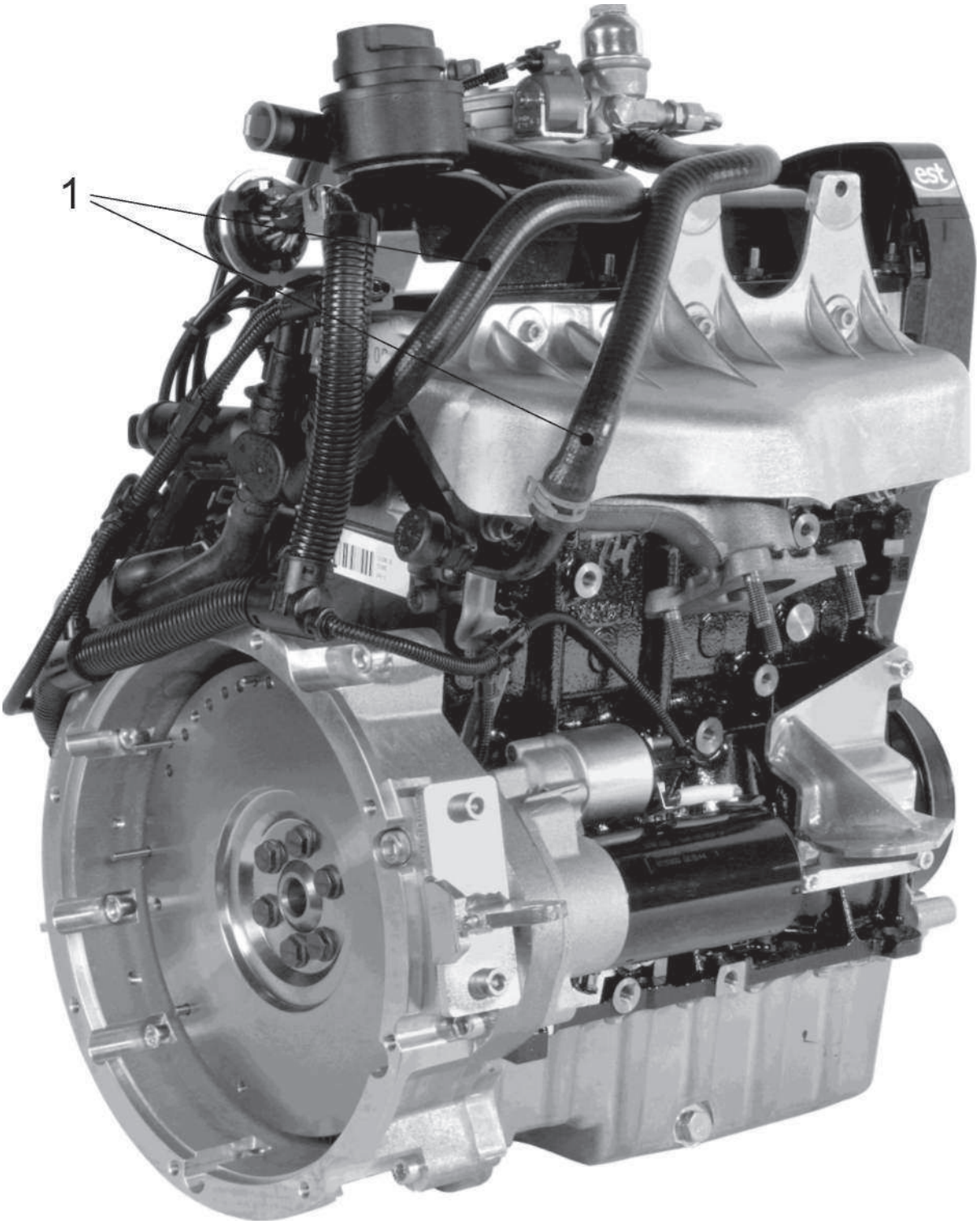
harness + mounting parts			
POS	**PART NUMBER**	**DESCRIPTION**	**QTY**
1	9 10 004	Bracket, 42p connector valve cover 2.0L	1
2	6 00 901	Harness, wiring 2.0L LPG	1
2	6 00 990	Harness, wiring 2.0L gasoline	1
-	6 60 900	Kit, connector housing receptacle 42p	
-	6 00 903	Harness contra	
-	6 00 904	Harness contra pressure regulator	

2.8. Starter + mounting parts



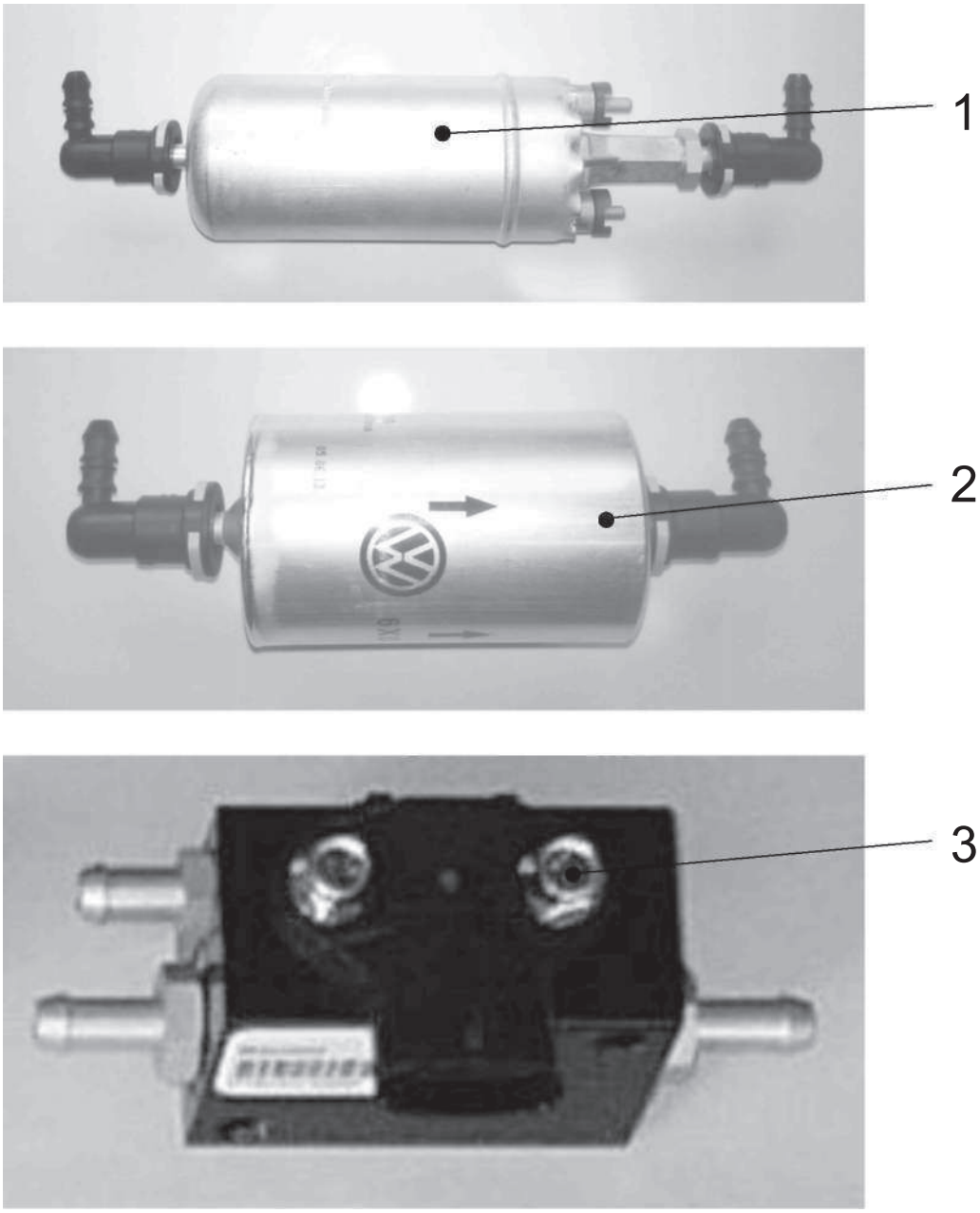
starter + mounting parts			
POS	**PART NUMBER**	**DESCRIPTION**	**QTY**
1	068 911 024G	starter	1
2	See kit below	Bolt, sockethead M12x100	1
3	See kit below	Bolt, sockethead M12x60	1
-	9 91 003	Kit, starter + mounting parts Kit contains: POS 1 + 2 + 3	1

2.9. Water hoses + mounting parts



Water hoses + mounting parts			
POS	**PART NUMBER**	**DESCRIPTION**	**QTY**
1	7 21 900	Kit, water hoses	1

2.10. Gasoline fuel parts



Gasoline fuel parts			
POS	**PART NUMBER**	**DESCRIPTION**	**QTY**
1	9 93 900	Kit, Fuel pump gasoline	1
2	6X0 121 511B	Kit, Fuel filter gasoline	1
3	9 93 902	Kit, Fuel pressure manifold	1